

EEBus UC Technical Specification

Control of Battery

Version 1.0.0

Cologne, 2023-12-31

EEBus Initiative e.V.

Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

Terms of use for publications of EEBus Initiative e.V.**General information**

The specifications, particulars, documents, publications and other information provided by the EEBus Initiative e.V. are solely for general informational purposes. Particularly specifications that have not been submitted to national or international standardisation organisations by EEBus Initiative e.V. (such as DKE/DIN-VDE or IEC/CENELEC/ETSI) are versions that have not yet undergone complete testing and can therefore only be considered as preliminary information. Even versions that have already been published via standardisation organisations can contain errors and will undergo further improvements and updates in future.

Liability

EEBus Initiative e.V. does not assume liability or provide a guarantee for the accuracy, completeness or up-to-date status of any specifications, data, documents, publications or other information provided and particularly for the functionality of any developments based on the above.

Copyright, rights of use and exploitation

The specifications provided are protected by copyright. Parts of the specifications have been submitted to national or international standardisation organisations by EEBus Initiative e.V., such as DKE/DIN-VDE or IEC/CENELEC/ETSI, etc. Furthermore, all rights to use and/or exploit the specifications belong to the EEBus Initiative e.V., Deutz-Mülheimer-Straße 183, 51063 Cologne, Germany and can be used in accordance with the following regulations:

The use of the specifications for informational purposes is allowed. It is therefore permitted to use information evident from the contents of the specifications. In particular, the user is permitted to offer products, developments and/or services based on the specifications.

Any respective use relating to standardisation measures by the user or third parties is prohibited. In fact, the specifications may only be used by EEBus Initiative e.V. for standardisation purposes. The same applies to their use within the framework of alliances and/or cooperations that pursue the aim of determining uniform standards.

Any use not in accordance with the purpose intended by EEBus Initiative e.V. is also prohibited.

Furthermore, it is prohibited to edit, change or falsify the content of the specifications. The dissemination of the specifications in a changed, revised or falsified form is also prohibited. The same applies to the publication of extracts if they distort the literal meaning of the specifications as a whole.

It is prohibited to pass on the specifications to third parties without reference to these rights of use and exploitation.

It is also prohibited to pass on the specifications to third parties without informing them of the authorship or source.

Without the prior consent of EEBus Initiative e.V., all forms of use and exploitation not explicitly stated above are prohibited.

Table of contents

Table of contents.....	3
List of figures	4
List of tables	5
1 Scope of the document	7
1.1 References.....	7
1.1.1 EEBUS documents	7
1.1.2 Normative references.....	7
1.2 Terms and definitions.....	7
1.3 Requirements	8
1.3.1 Requirements wording.....	8
1.3.2 Mapping of High-Level requirements.....	9
2 High-Level description.....	10
2.1 Introduction.....	10
2.1.1 Control mode "Power"	10
2.1.2 Control mode "PCC"	11
2.2 Detailed background information	12
2.2.1 Inverter device-hierarchy	12
2.2.2 Control mode details.....	13
2.3 System behaviour rules.....	14
2.4 State machine.....	18
2.4.1 Diagram	18
2.4.2 Defined states.....	19
2.4.3 Possible transitions.....	20
2.5 Actors	22
2.5.1 CEM	22
2.5.2 Inverter.....	22
2.6 Scenarios	23
2.6.1 Scenario 1 - Control mode "Power"	24
2.6.2 Scenario 2 - Control mode "PCC"	25
2.6.3 Scenario 3 - Configuration parameters	27
2.6.4 Scenario 4 - Failsafe values.....	29
2.6.5 Scenario 5 - Heartbeat.....	30
2.7 Dependencies to other Use Cases.....	31
2.7.1 "Limitation of Power Consumption"	31
2.7.2 "Limitation of Power Production"	32
2.7.3 "Monitoring of Inverter"	32
2.7.4 "Monitoring of Battery"	33
2.7.5 "Monitoring of PV String"	33
2.7.6 Monitoring of Power Consumption.....	33
2.7.7 "Monitoring of Grid Connection Point"	34
2.8 Further information and rules.....	34
2.8.1 Sign conventions.....	34
2.9 Assumptions and Prerequisites.....	35
3 Technical SPINE solution	36
3.1 General rules and information	36

47	3.1.1	Underlying technology documents	36
48	3.1.2	Use Case discovery rules	36
49	3.1.3	Rules for "Content of Specialization..." tables and "Content of Function..." tables	36
50	3.1.4	Rules for "Feature Types and Functions..." tables	46
51	3.1.5	"Actor ... overview" diagram rules	46
52	3.1.6	Specializations	47
53	3.1.7	Order of messages within the sequence diagrams	48
54	3.1.8	Further information and rules	48
55	3.2	Actors	49
56	3.2.1	CEM	49
57	3.2.2	Inverter	66
58	3.3	Pre-Scenario communication	81
59	3.3.1	General information	81
60	3.3.2	Detailed discovery	82
61	3.3.3	Binding	84
62	3.3.4	Subscription	84
63	3.3.5	Dynamic behaviour	85
64	3.4	Scenarios	85
65	3.4.1	Scenario 1 - Control mode "Power"	85
66	3.4.2	Scenario 2 - Control mode "PCC"	90
67	3.4.3	Scenario 3 - Configuration parameters	96
68	3.4.4	Scenario 4 - Failsafe values	100
69	3.4.5	Scenario 5 - Heartbeat	104
70			

71 List of figures

72	Figure 1: High-Level Use Case functionality overview	10
73	Figure 2: Example for Inverter device-hierarchy	12
74	Figure 3: COB state machine of Actor Inverter	19
75	Figure 4: Scenario overview	23
76	Figure 5: Actor overview example	47
77	Figure 6: Actor "CEM" overview	49
78	Figure 7: Actor "Inverter" overview	67
79	Figure 8: Pre-Scenario communication - Detailed discovery sequence diagram	83
80	Figure 9: Pre-Scenario communication - Binding sequence diagram	84
81	Figure 10: Pre-Scenario communication - Subscription sequence diagram	85
82	Figure 11: Scenario 1 - Initial Scenario communication sequence diagram	86
83	Figure 12: Scenario 1 - Runtime Scenario communication sequence diagram	88
84	Figure 13: Scenario 2 - Initial Scenario communication sequence diagram	91
85	Figure 14: Scenario 2 - Runtime Scenario communication sequence diagram	94
86	Figure 15: Scenario 3 - Initial Scenario communication sequence diagram	97
87	Figure 16: Scenario 3 - Runtime Scenario communication sequence diagram	99
88	Figure 17: Scenario 4 - Initial Scenario communication sequence diagram	101
89	Figure 18: Scenario 4 - Runtime Scenario communication sequence diagram	103
90	Figure 19: Scenario 5 - Initial Scenario communication sequence diagram	105

91	Figure 20: Scenario 5 - Runtime Scenario communication sequence diagram.....	106
92		

93 List of tables

94	Table 1: Inverter control behaviour	15
95	Table 2: Scenario implementation requirements for Actors	23
96	Table 3: Scenario 1 - Control mode "Power" - Data point list.....	25
97	Table 4: Scenario 2 - Control mode "PCC" - Data point list.....	27
98	Table 5: Scenario 3 - Configuration parameters - Data point list.....	28
99	Table 6: Scenario 4 - Failsafe values - Data point list	30
100	Table 7: Scenario 5 - Heartbeat - Data point list	31
101	Table 8: Presence indication description	37
102	Table 9: Example table for cardinality indications on Elements and list items.....	39
103	Table 10: Content of an example Specialization	43
104	Table 11: Presence indication of Feature Types and Functions support	46
105	Table 12: Specialization support for Actor CEM.....	50
106	Table 13: Feature Types and Functions used within this Use Case by the Actor CEM	51
107	Table 14: Content of Function "deviceDiagnosisHeartbeatData" at Actor CEM	52
108	Table 15: Content of Specialization "Setpoint_BatteryAcPower" at Actor CEM	54
109	Table 16: Content of Specialization "Setpoint_BatteryDcPower" at Actor CEM	55
110	Table 17: Content of Specialization "Setpoint_PccPower" at Actor CEM.....	56
111	Table 18: Content of Specialization "DeviceConfiguration_MaxAcChargePower" at Actor CEM.....	57
112	Table 19: Content of Specialization "DeviceConfiguration_MaxAcDischargePower" at Actor CEM	58
113	Table 20: Content of Specialization "DeviceConfiguration_MaxDcChargePower" at Actor CEM	59
114	Table 21: Content of Specialization "DeviceConfiguration_MaxDcDischargePower" at Actor CEM	59
115	Table 22: Content of Specialization "DeviceConfiguration_BatteryActiveControlMode" at Actor CEM	
116		60
117	Table 23: Content of Specialization "DeviceConfiguration_DefaultAcPower" at Actor CEM	61
118	Table 24: Content of Specialization "DeviceConfiguration_DefaultDcPower" at Actor CEM	62
119	Table 25: Content of Specialization "DeviceConfiguration_DefaultPccPower" at Actor CEM.....	62
120	Table 26: Content of Specialization "DeviceConfiguration_FailsafeAcPowerSetpoint" at Actor CEM .	63
121	Table 27: Content of Specialization "DeviceConfiguration_FailsafeDcPowerSetpoint" at Actor CEM .	64
122	Table 28: Content of Specialization "DeviceConfiguration_FailsafePccPowerSetpoint" at Actor CEM	65
123	Table 29: Content of Specialization "DeviceConfiguration_FailsafeDurationMinimum" at Actor CEM	65
124	Table 30: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor CEM .	66
125	Table 31: Specialization support for Actor Inverter	69
126	Table 32: Feature Types and Functions used within this Use Case by the Actor Inverter.....	69
127	Table 33: Content of Function "setpointDescriptionListData" at Actor Inverter.....	70
128	Table 34: Content of Function "setpointConstraintsListData" at Actor Inverter	72
129	Table 35: Content of Function "setpointListData" at Actor Inverter	73
130	Table 36: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Inverter	75
131	Table 37: Content of Function "deviceConfigurationKeyValueListData" at Actor Inverter	79
132	Table 38: Content of Function "deviceDiagnosisHeartbeatData" at Actor Inverter	80
133	Table 39: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor Inverter	
134		81

135	Table 40: Initial Scenario communication content references for Scenario 1	87
136	Table 41: Runtime Scenario communication content references for Scenario 1	89
137	Table 42: Initial Scenario communication content references for Scenario 2	93
138	Table 43: Runtime Scenario communication content references for Scenario 2	95
139	Table 44: Initial Scenario communication content references for Scenario 3	98
140	Table 45: Runtime Scenario communication content references for Scenario 3	100
141	Table 46: Initial Scenario communication content references for Scenario 4	102
142	Table 47: Runtime Scenario communication content references for Scenario 3	104
143	Table 48: Initial Scenario communication content references for Scenario 5	105
144	Table 49: Runtime Scenario communication content references for Scenario 5	106
145		
146		

1 Scope of the document

This document describes the Use Case "Control of Battery" (short-name: COB). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or an electric vehicle).

Battery Inverter

Inverter that has only batteries as DC input.

COB

Control of Battery (short name of this Use Case)

Configuration Parameter

Each data point of Scenario 3 is called a "Configuration Parameter" (see [COB-031] to [COB-034]).

Control Mode Setpoint

The setpoints Charge/Discharge AC Power Setpoint ([COB-011]), Charge/Discharge DC Power Setpoint ([COB-012]) and PCC Power Setpoint ([COB-021]) are all called Control Mode Setpoint.

Default Setpoint

The setpoints Default AC Power ([COB-032]), Default DC Power ([COB-033]) and Default PCC Power ([COB-034]) are all called Default Setpoint.

Failsafe Setpoint

The setpoints Failsafe Charge/Discharge AC Power Setpoint ([COB-041]), Failsafe Charge/Discharge DC Power Setpoint ([COB-042]) and Failsafe PCC Power Setpoint ([COB-043]) are all called Failsafe Setpoint.

182 Failsafe Value

183 Each data point of Scenario 4 is called a "Failsafe Value" (see [COB-041] to [COB-044]).

184 GCP

185 Abbreviation for Grid Connection Point.

186 Hybrid Inverter

187 Inverter that has batteries and PV panels as DC input.

188 MGCP

189 Short name of the Use Case "Monitoring of Grid Connection Point"

190 MOB

191 Short name of the Use Case "Monitoring of Battery"

192 MOI

193 Short name of the Use Case "Monitoring of Inverter"

194 MPC

195 Short name of the Use Case "Monitoring of Power Consumption"

196 MPS

197 Short name of the Use Case "Monitoring of PV String"

198 PCC

199 Abbreviation for Point of Common Coupling.

200 Photovoltaic Inverter

201 Inverter that has only PV panels as DC input.

202 Scenario

203 Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more
204 quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or
205 optional.

206 Setpoint

207 Any of Control Mode Setpoint, Default Setpoint or Failsafe Setpoint.

208 Specialization

209 Reusable data collection for a specific functionality.

210 SPINE

211 Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative
212 e.V.

213

214 1.3 Requirements**215 1.3.1 Requirements wording**

216 The following keywords are used:

- 217 - SHALL
- 218 - SHALL NOT
- 219 - SHOULD
- 220 - SHOULD NOT
- 221 - MAY

222 Note: They apply only if written in capital letters.

223 For the meaning of the keywords, please refer to [RFC2119].

224

225 **1.3.2 Mapping of High-Level requirements**

226 Within the High-Level Use Case description, the following abbreviation is used:

227 [COB-xyz]

228 e.g.: [COB-007]

229 The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique
230 number xyz. These requirements are referenced throughout the technical solution to show how each
231 High-Level requirement is realized in the technical part.

232

2 High-Level description

2.1 Introduction

Today, battery systems independently control the grid connection point with the aim of drawing no or a defined power from the grid. In the extended energy management context, energy managers are used to harmonize energy demand and generation. The aim of energy management is to control energy flows within the property in such a way that the devices are operated in a price- or CO₂-optimized way, e.g. by using PV energy. The energy manager knows the requirements of the connected devices, the base load of a building and learns consumption profiles. Together with PV forecasts, the energy manager calculates the consumption plans and optimizes them over the day, if conditions in the system change.

In order to be able to integrate battery systems more efficiently into energy management systems, it must be possible to control the operating behaviour of the battery system. A pure control to the grid connection point might be counterproductive for some applications.

It must be possible to influence the operating behaviour as follows:

1. Control mode "Power"
2. Control mode "PCC" (Point of Common Coupling)

It shall be possible to select the respective operating mode using the "Active Control Mode" ([COB-031]) parameter.

A Heartbeat mechanism is used to identify communication problems or other failures of the two communication partners. For this, both Actors send a cyclic Heartbeat to the respective counterpart (at least every 60 seconds). A missing of Heartbeats for more than 120 seconds indicates a communication problem. In such a case, the Inverter goes into a failsafe mode where configured setpoints must be followed (as good as possible). This failsafe mechanism is intended to avoid uncontrolled feed-in power in case of connectivity problems or when the Inverter is repowered for any reason.

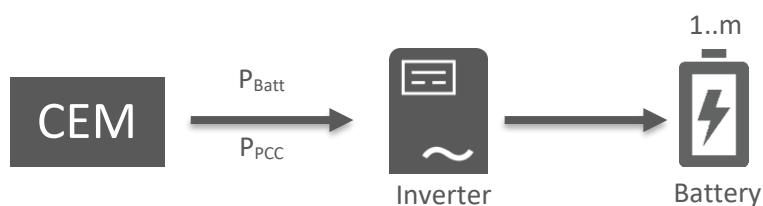


Figure 1: High-Level Use Case functionality overview

Note: This Use Case is intended to control the Inverter's battery system and not single batteries that are attached to the Inverter.

2.1.1 Control mode "Power"

The setpoint P for charging and discharging power is used to control the Inverter's battery or batteries ([COB-011] for AC power setpoint, [COB-012] for DC power setpoint).

266 With this control mode, only the combined power consumption or production of the attached
267 batteries is controlled, not any power provided by PV strings.

268 It must be possible to specify the charge/discharge power as value P_{Batt} in Watt. The sign indicates
269 whether the setpoint value is a charge or discharge power setpoint. According to the "passive sign
270 convention", the charging power SHALL be indicated by a positive sign and the discharging power by
271 a negative sign (see section 2.8.1 for details).

272 Charging and discharging power values are carried out either as AC or DC value. For an Inverter that
273 has only batteries connected, the AC value SHALL be used as the setpoint value ([COB-011]), whereas
274 for hybrid systems (PV/battery) the DC value SHALL be used as the setpoint value ([COB-012]).

275 Note: In control mode "Power" the energy manager is responsible for compliance with the feed-in
276 regulations, e.g. in the case of Germany the FNN regulation.

277

278 **2.1.2 Control mode "PCC"**

279 **2.1.2.1 Setpoint PCC**

280 The PCC setpoint input is used to specify the power reference at the grid connection point ([COB-
281 021]). In this case, the Inverter regulates to the specified setpoint. This means that a PCC setpoint of
282 "0W" SHOULD lead to a zero feed-in and zero consumption at the grid connection point.

283 With this control mode, the combined power of the attached batteries and PV strings is used to
284 follow the given setpoint at the grid connection point.

285 The control mode "PCC" can react faster to changing situations than the control mode "Power" as
286 the Inverter itself regulates its power according to a power measurement value at the grid
287 connection point (GCP), instead of an energy manager analysing the current situation and configuring
288 the Inverter accordingly.

289 It SHALL be possible to specify the power value at the grid connection point as value P_{PCC} in Watt.

290 The sign indicates whether the power is drawn from or delivered to the grid. According to the
291 "passive sign convention", drawn power SHALL be identified by a positive sign and delivered power
292 by a negative sign (see section 2.8.1 for details).

293 If the P_{PCC} setpoint exceeds the currently possible value (e.g., greater than the nominal power of the
294 Inverter (provided by the Use Case "Monitoring of Inverter", see section 2.7.3) or a negative value in
295 case of an empty battery), the Inverter SHALL still try to reach the PCC setpoint within the Inverter's
296 current power constraints. This rule applies for both directions (grid consumption and feed-in).

297

2.1.2.2 Maximum Charge/Discharge Power Limits

In PCC mode, the Inverter SHALL enable the CEM to configure the maximum charge/discharge power the Inverter SHALL NOT exceed for its connected batteries. The values SHALL be defined as absolute AC or DC values in Watt:

- Maximum AC charge power ([COB-022], for Battery Inverters)
- Maximum AC discharge power ([COB-023], for Battery Inverters)
- Maximum DC charge power ([COB-024], for Hybrid Inverters)
- Maximum DC discharge power ([COB-025], for Hybrid Inverters)

The CEM is not obliged to set the maximum AC/DC charge/discharge limits. A vendor SHALL pre-configure appropriate limits (e.g., set them to the nominal values).

The maximum AC/DC charge/discharge power limits of the Inverter SHALL be applied if and only if the Control Mode PCC is activated. If the limit values ([COB-022], [COB-023], [COB-024], [COB-025]) exceed the Inverter's maximum possible charge/discharge power, the limit values do not have any effect on the Inverter.

Depending on the system type (battery only or hybrid), the respective limits SHALL be available.

2.2 Detailed background information

2.2.1 Inverter device-hierarchy

This Use Case describes the controlling of the battery of an Inverter. Although the battery functionality is controlled, the Actor that is addressed by this Use Case still is the Inverter, not the battery (as opposed to the Use Case "Monitoring of Battery") as not a single battery is controlled but the aggregated battery functionality of the Inverter.

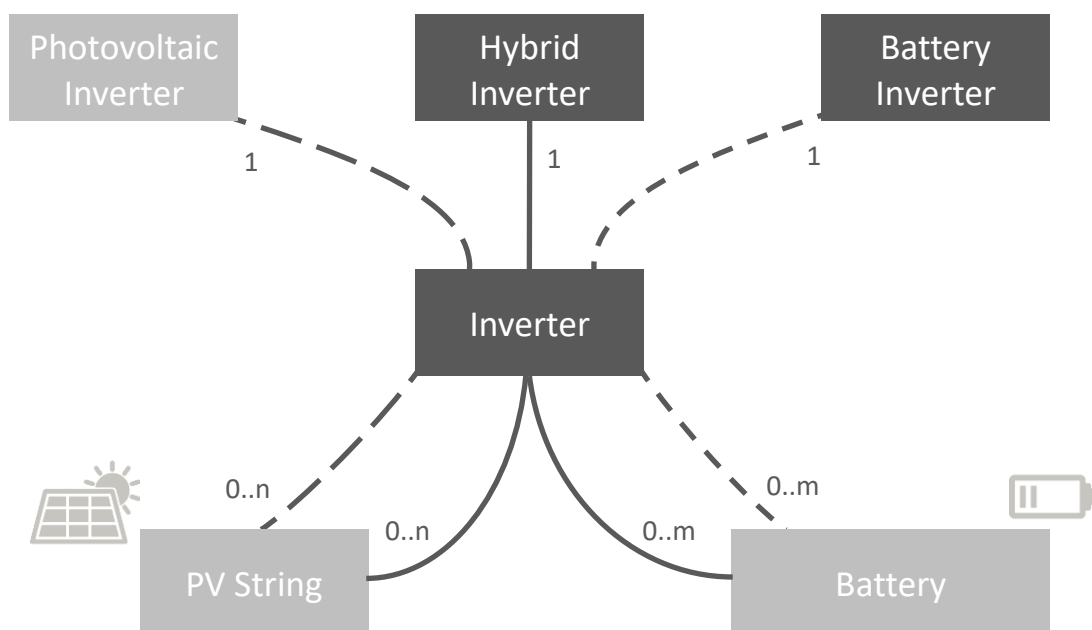


Figure 2: Example for Inverter device-hierarchy

The context of the Inverter is given by its so-called "parent" in the device-hierarchy. This can be:

- "Photovoltaic Inverter" ([COB-970/1]), meaning that the Actor Inverter has only PV strings (up to "n" in Figure 2) as so-called "children" in its device hierarchy
- "Hybrid Inverter" ([COB-970/2]), meaning that the Actor Inverter is a hybrid Inverter that has both PV strings (up to "n" in Figure 2) and batteries (up to "m" in Figure 2) as "children" in its device hierarchy
- "Battery Inverter" ([COB-970/3]), meaning that the Actor Inverter has only batteries (up to "m" in Figure 2) as "child" in its device hierarchy.

Note: The "0" in "0..n" and "0..m" of Figure 2 just expresses that this Use Case does neither require nor describe data of any PV string or battery itself, as such information is covered by related Use Cases discussed in section 2.7. Instead, the Use Case primarily focuses on the electrical functionality dedicated to the circuit for attached batteries, where PV strings may still contribute to the control mode "PCC".

2.2.2 Control mode details

2.2.2.1 Control mode relations

The following control modes are defined:

- Power [COB-031/1], see section 2.1.1
- PCC [COB-031/2], see section 2.1.2
- Auto [COB-031/3], see section 2.2.2.2

The Actor CEM can change the control mode in the following ways:

- Power → PCC *¹
- PCC → Power *¹
- Auto → Power *²
- Auto → PCC *²

*¹: The Actor Inverter might not always accept a change on the control mode (see section 2.3 for details).

*²: There may be situations where a change from control mode "Auto" to another control mode triggered by the Actor CEM is not possible (e.g. during trickle charging) (see section 2.6.3.1 for details).

The Actor Inverter can change the control mode in the following ways:

- Power → Auto
- PCC → Auto
- Auto → Power
- Auto → PCC

For rules regarding the control mode changes, please refer to section 2.3.

2.2.2.2 Control mode "Auto"

The control mode "Auto" is (e.g.) used for the following situations:

- The Inverter goes into a mode where a change by another instance is not allowed (e.g. during trickle charging).
- After a longer connection problem to the Actor CEM, where it might be suitable to not stay in Failsafe State but rather change to a mode where the Actor Inverter decides how to operate.

In control mode "Auto" the Inverter produces or consumes power as if no setpoints are set by the Inverter (in the context of this Use Case).

An installer or user shall configure the Inverter according to the local regulations and needed system functionality (e.g., if no energy manager is available, the Inverter regulates the PCC to "0W").

The Actor CEM is not allowed to change the control mode into the mode "Auto" (only the Actor Inverter may do so). The control mode "Auto" is triggered automatically when the Failsafe State is left after a certain period of time (see section 2.3).

2.3 System behaviour rules

The Inverter's power consumption/production is controlled by different datapoints:

- Control Mode Setpoints ([COB-011], [COB-012], [COB-021])
- Default Setpoints ([COB-032], [COB-033], [COB-034])
- Failsafe Setpoints ([COB-041], [COB-042], [COB-043]).

Which Setpoint is used depends on the type of the Inverter (see sections 2.2.1, 2.6.1.1, 2.6.2.1, 2.6.3.1, 2.6.4.1), the current Active Control Mode ([COB-031]), the according Control Mode Setpoint as well as the Inverter's state (see section 2.4).

Definitions:

- "Evaluate the ability or inability to apply the Control Mode Setpoint" means the Inverter evaluates if the Control Mode Setpoint is appropriate in the event that the respective control mode is already activated or may become activated.
- "Apply Setpoint" means the Inverter will use the (Control Mode or Default) Setpoint if and only if the respective control mode is activated.

Note: Applying a Setpoint does NOT include an automatic activation of the respective control mode. A control mode is only activated where stated explicitly. Setting a Control Mode Setpoint to "activated" does also not automatically include the activation of the control mode.

Depending on the status of the Heartbeat and the Control Mode Setpoints ([COB-011], [COB-012] and [COB-021]) the following behaviour can be expected at the Inverter for the respective referenced states (further states will be explained below):

Heartbeat	Control Mode Setpoint	Inverter behaviour	Applied in state
Received in time ([COB-900/1])	Deactivated	Active Control Mode dependent Default Setpoint applied	- Power Control State - PCC Control State

	Was activated, but duration expired	Active Control Mode dependent Default Setpoint applied	
	Activated (duration not expired / no duration defined)	Inverter controlled by Active Control Mode dependent Control Mode Setpoint	
Not received in time ([COB-900/2])	Don't care	Controlled by Active Control Mode dependent Failsafe Setpoint	- Failsafe State

Table 1: Inverter control behaviour

Whether a Heartbeat is received in time ([COB-900/1]) or is not received in time ([COB-900/2]) can be seen for the respective conditions throughout this specification.

After any kind of restart of the Inverter, the Inverter starts in state "Init" (see section 2.4.2) and SHALL begin with a consumption or production as stated in the Failsafe Setpoint*¹ (see section 2.6.4) according to the Active Control Mode*² ([COB-901]). All Control Mode Setpoints SHALL be deactivated ([COB-009/2]).

*¹: If there was no change of the according Failsafe Setpoint ([COB-041], [COB-042] or [COB-043]) by the CEM before restart or if the earlier written data was lost during restart, the Inverter SHALL use its initially pre-configured Failsafe Setpoint ([COB-902]).

*²: If the Active Control Mode was not set before or if the last Active Control Mode was not stored persistently, the Inverter SHALL set the Active Control Mode to "PCC", hence the Inverter's power consumption or production SHALL follow the according Failsafe Setpoint for the Active Control Mode "PCC" ([COB-903]).

Note: The Active Control Mode's value and the Inverter's state do not correlate in every case. E.g., as defined above, the Active Control Mode may be set to "Power", but the Inverter is still in state "Init". The rules when to change the state are defined within this section.

In state "Init", only after a CEM's Heartbeat and a following first write command on the Active Control Mode ([COB-031]) is received and accepted by the Inverter, the Inverter switches either to state "Power Control State" ([COB-904/1]) or "PCC Control State" ([COB-904/2]) (see section 2.4.2 for details). If the received Active Control Mode is not accepted by the Inverter, the Inverter SHALL switch to "Auto Control State" ([COB-904/3]).

If the Inverter does not receive any Heartbeat or receives a Heartbeat but no following write on the Active Control Mode from the CEM within 120 seconds since entering the state "Init", the Inverter MAY switch into state "Auto Uncontrolled State" (see section 2.4.2) ([COB-905]).

Upon receipt of a Control Mode Setpoint the Inverter SHALL evaluate its ability or inability to apply the Control Mode Setpoint. If the Control Mode Setpoint can be applied, the Inverter SHALL inform the CEM that the Control Mode Setpoint is accepted (ACK, [COB-002/1]), otherwise the Inverter SHALL inform the CEM that the Control Mode Setpoint is rejected (NACK, [COB-003/1]).

Upon receipt of a Configuration Parameter the Inverter SHALL evaluate its ability or inability to apply the Configuration Parameter. If the Configuration Parameter can be applied, the Inverter SHALL

424 inform the CEM that the Configuration Parameter is accepted (ACK, [COB-002/2]), otherwise the
425 Inverter SHALL inform the CEM that the Configuration Parameter is rejected (NACK, [COB-003/2]).

426 Upon receipt of a Failsafe Value the Inverter SHALL evaluate its ability or inability to apply the
427 Failsafe Value. If the Failsafe Value can be applied, the Inverter SHALL inform the CEM that the
428 Failsafe Value is accepted (ACK, [COB-002/3]), otherwise the Inverter SHALL inform the CEM that the
429 Failsafe Value is rejected (NACK, [COB-003/3]).

430 If the Inverter is neither in state "Init" nor "Failsafe State" and rejects the write command on the
431 Active Control Mode, the Inverter SHALL stay in the current Active Control Mode ([COB-906]).

432 If the Inverter rejects the write command on a Setpoint, the Inverter's Setpoint value remains
433 unchanged.

434 If the Inverter receives a Control Mode Setpoint that is not used within the current Active Control
435 Mode, the Inverter SHALL (if it accepts the value) store it for later usage (when the according control
436 mode becomes active).

437 The following rules specify the Inverter's behaviour on written Setpoints ([COB-960]):

- 438 - In general, an Inverter SHOULD accept all Setpoints that are within the technically possible
439 range of the Inverter.
- 440 - An Inverter MAY accept a Setpoint that contradicts a limit set by a Use Case with a higher
441 priority than this Use Case (see [COB-951], [COB-952] and [COB-953] in section 2.7 for details
442 on these Use Cases), although the Inverter will not follow the Setpoint at this time (but as
443 soon as the limit with the higher priority is deactivated or changed to a higher value, the
444 Inverter SHALL follow this Setpoint, within its current capabilities).
- 445 - An Inverter MAY accept a Setpoint even though it can temporarily not fulfil it due to
446 regulatory specifications (as soon as this regulatory limitation does not apply anymore, the
447 Inverter SHALL then follow the given Setpoint, within its current capabilities).
- 448 - An Inverter SHALL reject a Setpoint that contradicts permanent regulatory specifications.
- 449 - The Inverter MAY reject a Setpoint that contradicts the current Inverter's constraints for the
450 according Setpoint (see [COB-980/1] and [COB-980/2]). If the Inverter accepts the Setpoint
451 nevertheless, the Inverter will still consider its internal constraints and therefore will not
452 meet the Setpoint exactly.
- 453 - An Inverter SHALL reject a Setpoint that is technically not possible to follow (i.e., outside the
454 technical specifications of the appliance).
- 455 - An Inverter SHALL reject a Setpoint for the Active Control Mode "Power" that exceeds the
456 Inverter's nominal values.

457 Note: If in state "Power Control State", "PCC Control State" or "Auto Control State", the CEM may
458 also write a Setpoint that belongs to a control mode other than the currently Active Control Mode.

459 Note: A CEM should be aware of the possibility of its write commands being rejected. Appropriate
460 reactions on those NACK messages should be implemented (e.g. retry at a later time or change the
461 chosen value).

462 If within state "Power Control State" the Inverter receives and accepts a write command on the
463 Active Control Mode with content "PCC", the Inverter SHALL switch to state "PCC Control State"
464 ([COB-908]). If the command is rejected, [COB-906] holds valid.

465 If within state "PCC Control State" the Inverter receives and accepts a write command on the Active
466 Control Mode with content "Power", the Inverter SHALL switch to state "Power Control State" ([COB-
467 909]). If the command is rejected, [COB-906] holds valid.

468 If in state "Power Control State" or "PCC Control State" the Inverter MAY change its state to "Auto
469 Control State" ([COB-910/1]) by changing the Active Control Mode to "Auto" ([COB-031/3]). When
470 such an autonomous state change is performed, is up to the business logic of the Inverter.

471 If the Actor Inverter changes its control mode from "Power" or "PCC" to "Auto" for some reason, the
472 Inverter SHOULD switch back to the according control mode ("Power" or "PCC") if the control mode
473 "Auto" is not needed anymore.

474 Note: When switching to "Auto Control State", all currently activated Setpoints keep being activated
475 (in contrast to the states "Init" and "Failsafe State" where all Setpoints are deactivated).

476 If within state "Auto Control State" the Inverter receives and accepts a write command on the Active
477 Control Mode with content "Power", the Inverter SHALL switch to state "Power Control State" ([COB-
478 911]). If the command is rejected, [COB-906] holds valid.

479 If within state "Auto Control State" the Inverter receives and accepts a write command on the Active
480 Control Mode with content "PCC", the Inverter SHALL switch to state "PCC Control State" ([COB-
481 912]). If the command is rejected, [COB-906] holds valid.

482 If in state "Auto Control State" the Inverter MAY change its state to "Power Control State" or "PCC
483 Control State" ([COB-910/2]) by changing the Active Control Mode to "Power" ([COB-031/1]) or
484 "PCC" ([COB-031/2]). When such an autonomous state change is performed, is up to the business
485 logic of the Inverter.

486 The Actor Inverter itself SHOULD only change the control mode in important cases.

487 If in state "Power Control State", "PCC Control State" or "Auto Control State", the Inverter SHALL
488 switch into "Failsafe State" if there is no Heartbeat received from the CEM for at least 120 seconds
489 since the last Heartbeat ([COB-913]). All Control Mode Setpoints SHALL then be deactivated ([COB-
490 009/2]).

491 After restoration of communication, the CEM SHALL send a Heartbeat and a following Active Control
492 Mode within 60 seconds to the Inverter after having determined that the communication is possible
493 again.

494 In state "Init" or "Failsafe State" or state "Auto Uncontrolled State", only after a Heartbeat from the
495 CEM, a following received write command within 60 seconds on the Active Control Mode SHALL be
496 evaluated. Earliest thereafter, write commands on any other data point defined in this Use Case
497 SHALL be evaluated by the Inverter.

498 In "Failsafe State", the Inverter SHALL leave the state upon receipt of a Heartbeat from the CEM and
499 a following write on the Active Control Mode*³, even if the write command will not be accepted
500 (resulting in a NACK according to [COB-003/2])*⁴.

501 *³: If the write command on the Active Control Mode is accepted by the Inverter and the content of
502 the Active Control Mode write command is "Power", the Inverter SHALL switch into state "Power
503 Control State" ([COB-914/1]). If the write command on the Active Control Mode is accepted by the
504 Inverter and the content of the Active Control Mode write command is "PCC", the Inverter SHALL
505 switch into state "PCC Control State" ([COB-914/2]).

506 *⁴: In this case, the Inverter SHALL switch to "Auto Control State" ([COB-914/3]).

507 The Inverter MAY leave the "Failsafe State" and switch into state "Auto Uncontrolled State" if the
508 CEM's Heartbeat is received again, but no write command on the Active Control Mode is received
509 within 120 seconds ([COB-915]). This rule ensures that the Inverter is not blocked in "Failsafe State"
510 by a non-functional CEM.

511 The Inverter MAY leave the "Failsafe State" after expiry of the Failsafe Duration Minimum (see [COB-
512 044]) and switch into state "Auto Uncontrolled State" ([COB-916]).

513 The CEM SHOULD monitor the actual AC (for pure battery inverters) or DC (for hybrid inverters)
514 power consumption or production of the Inverter as well as the actual power at the GCP (with the
515 Use Case "Monitoring of Grid Connection Point" (see section 2.7.7)). The Inverter SHOULD provide its
516 actual values for the AC (for pure battery inverters) or DC (for hybrid inverters) power using the Use
517 Case "Monitoring of Inverter" (see section 2.7.3) for providing the AC value and the Use Case
518 "Monitoring of Battery" (see section 2.7.4) for providing the DC value.

519 If the Inverter has no activated Control Mode Setpoint for the current Active Control Mode, the
520 according Default Setpoint SHALL be applied ([COB-917]):

- 521 - In control mode "Power":
 - 522 ○ For battery Inverters: "Default AC Power" ([COB-032]) setpoint is used
 - 523 ○ For hybrid Inverters: "Default DC Power" ([COB-033]) setpoint is used
- 524 - In control mode "PCC": "Default PCC Power" ([COB-034]) setpoint is used.

525

526 **2.4 State machine**

527 **2.4.1 Diagram**

528 The following state machine diagram shows how the Inverter switches between its states.

529 Figure 3 only shows the defined states and the possible transitions. Descriptions on the states can be
530 found in section 2.4.2, descriptions on the possible transitions can be found in section 2.4.3.

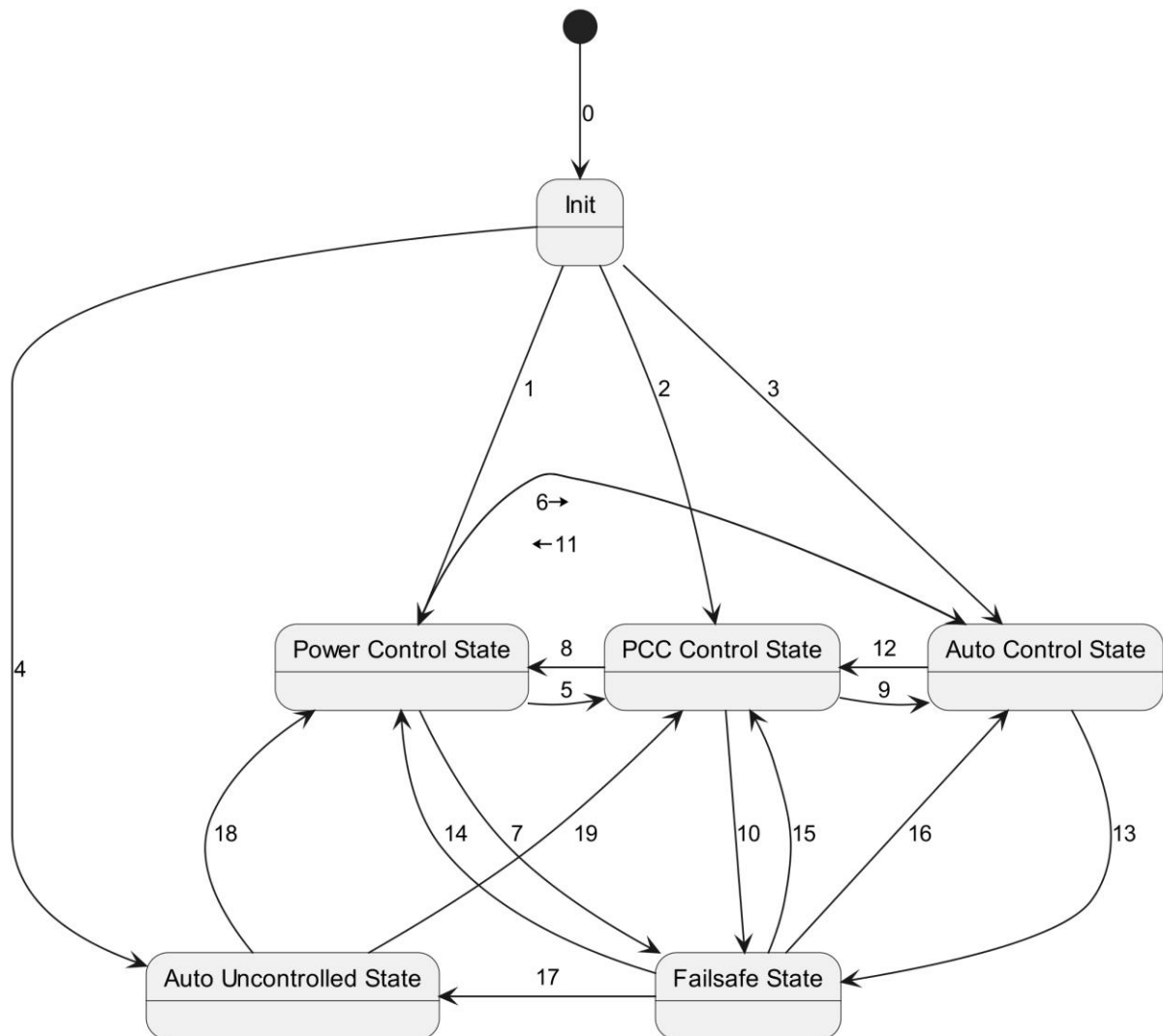


Figure 3: COB state machine of Actor Inverter

Note: The COB's state machine is independent from the ones specified in the Use Cases "Limitation of Power Consumption" and "Limitation of Power Production" (see sections 2.7.1 and 2.7.2 for details).

2.4.2 Defined states

Init: Inverter starts in "Init" state after completion of its (re)start; Inverter's power consumption or production controlled by the Failsafe Setpoints according to [COB-901]. All Control Mode Setpoints SHALL be deactivated ([COB-009/2]).

Power Control State: The Active Control Mode is set to "Power". The Heartbeats are received in time. The Inverter is controlled by:

- For battery Inverters:
 - Charge/Discharge AC Power Setpoint ([COB-011]) or
 - if [COB-011] is deactivated: Default AC Power ([COB-032])
- For hybrid Inverter:
 - Charge/Discharge DC Power Setpoint ([COB-012]) or

548 ○ if [COB-012] is deactivated: Default DC Power ([COB-033]).

549 All Control Mode Setpoints remain activated or deactivated (hence, their activation state does not
550 change).

551

552 **PCC Control State:** The Active Control Mode is set to "PCC". The Heartbeats are received in time. The
553 Inverter is controlled by the PCC Power Setpoint ([COB-021]). If [COB-021] is deactivated, the Inverter
554 is controlled by Default PCC Power ([COB-034]). All Control Mode Setpoints remain activated or
555 deactivated (hence, their activation state does not change).

556 **Auto Control State:** The Active Control Mode is set to "Auto". The Heartbeats are received in time.
557 The Inverter is not controlled by any Control Mode Setpoint or Default Setpoint. The power
558 consumption or production of the Inverter depends on internal business logic of the Inverter. All
559 Control Mode Setpoints remain activated or deactivated (hence, their activation state does not
560 change).

561 **Auto Uncontrolled State:** The Active Control Mode is set to "Auto". There are no Heartbeats from
562 the CEM. The Inverter is not controlled by any Control Mode Setpoint or Default Setpoint. The power
563 consumption or production of the Inverter depends on internal business logic of the Inverter.

564 **Failsafe State:** The Active Control Mode remains at the last value, unless the Inverter needs to adjust
565 the Active Control Mode autonomously while being in Failsafe State. There are no Heartbeats from
566 the CEM. The Inverter is not controlled by the CEM, but is limited by the Failsafe Setpoints
567 (depending on the Active Control Mode). All Control Mode Setpoints SHALL be deactivated ([COB-
568 009/2]).

569

570 **2.4.3 Possible transitions**

571 **0: ● --> Init:** Restart of Inverter completed. Applied rules: [COB-901], [COB-902], [COB-903].

572

573 **1: Init --> Power Control State:** Heartbeat and a following write on the Active Control Mode with the
574 value "Power" received and accepted within 120 seconds. The Default Setpoint (according to the
575 type of the Inverter) is applied until the respective Control Mode Setpoint is activated by the CEM
576 and accepted by the Inverter (see section 2.3 for details). Applied rules: [COB-904/1], [COB-917].

577 **2: Init --> PCC Control State:** Heartbeat and a following write on the Active Control Mode with the
578 value "PCC" received and accepted within 120 seconds. The according Default Setpoint is applied
579 until the respective Control Mode Setpoint is activated by the CEM and accepted by the Inverter (see
580 section 2.3 for details). Applied rules: [COB-904/2], [COB-917].

581 **3: Init --> Auto Control State:** Heartbeat and a following write on the Active Control Mode received
582 within 120 seconds that is not accepted by the Inverter. Applied rules: [COB-904/3].

583 **4: Init --> Auto Uncontrolled State:** The Inverter does not receive any Heartbeat or receives a
584 Heartbeat but no following write on the Active Control Mode from the CEM within 120 seconds since
585 entering the state "Init". Applied rules: [COB-905].

586

587 **5: Power Control State --> PCC Control State:** A write on the Active Control Mode with the value
588 "PCC" received and accepted (Heartbeat received in time ([COB-900/1])). If the PCC Power Setpoint is
589 activated, it is applied. Otherwise, the Default PCC Power is applied. Applied rules: [COB-908], [COB-
590 917].

591 **6: Power Control State --> Auto Control State:** Autonomous state change of the Inverter (Heartbeat
592 received in time ([COB-900/1])). Applied rules: [COB-910/1].

593 **7: Power Control State --> Failsafe State:** No Heartbeat received within 120 seconds since the last
594 Heartbeat. Applied rules: [COB-913].

595

596 **8: PCC Control State --> Power Control State:** A write on the Active Control Mode with the value
597 "Power" received and accepted (Heartbeat received in time ([COB-900/1])). If either
598 Charge/Discharge AC Power Setpoint or Charge/Discharge DC Power Setpoint is activated, the
599 according setpoint is applied. Otherwise, the according Default Setpoint is applied (either Default AC
600 Power or Default DC Power). Applied rules: [COB-909], [COB-917].

601 **9: PCC Control State --> Auto Control State:** Autonomous state change of the Inverter (Heartbeat
602 received in time ([COB-900/1])). Applied rules: [COB-910/1].

603 **10: PCC Control --> Failsafe State:** No Heartbeat received within 120 seconds since the last
604 Heartbeat. Applied rules: [COB-913].

605

606 **11: Auto Control State --> Power Control State:** A write on the Active Control Mode with the value
607 "Power" received and accepted (Heartbeat received in time ([COB-900/1])) or an autonomous state
608 change of the Inverter. If either Charge/Discharge AC Power Setpoint or Charge/Discharge DC Power
609 Setpoint is activated, the according setpoint is applied. Otherwise, the according Default Setpoint is
610 applied (either Default AC Power or Default DC Power). Applied rules: [COB-910/2], [COB-911], [COB-
611 917].

612 **12: Auto Control State --> PCC Control State:** A write on the Active Control Mode with the value
613 "PCC" received and accepted (Heartbeat received in time ([COB-900/1])) or an autonomous state
614 change of the Inverter. If the PCC Power Setpoint is activated, it is applied. Otherwise, the Default
615 PCC Power is applied. Applied rules: [COB-910/2], [COB-912], [COB-917].

616 **13: Auto Control State --> Failsafe State:** No Heartbeat received within 120 seconds since the last
617 Heartbeat. Applied rules: [COB-913].

618

14: Failsafe State --> Power Control State: Heartbeat and a following write on the Active Control Mode with the value "Power" received and accepted. The Default Setpoint (according to the type of the Inverter) is applied until the respective Control Mode Setpoint is activated by the CEM and accepted by the Inverter (see section 2.3 for details). Applied rules: [COB-914/1], [COB-917].

15: Failsafe State --> PCC Control State: Heartbeat and a following write on the Active Control Mode with the value "PCC" received and accepted. The according Default Setpoint is applied until the respective Control Mode Setpoint is activated by the CEM and accepted by the Inverter (see section 2.3 for details). Applied rules: [COB-914/2], [COB-917].

16: Failsafe State --> Auto Control State: Heartbeat and a following write on the Active Control Mode (with a value of "Power" or "PCC") was received but not accepted by the Inverter. Applied rules: [COB-914/3].

17: Failsafe State --> Auto Uncontrolled State: The Failsafe Duration Minimum expires or the Inverter receives a Heartbeat but no write on the Active Control Mode within 120 seconds. Applied rules: [COB-915], [COB-916].

18: Auto Uncontrolled State --> Power Control State: Heartbeat and a following write on the Active Control Mode with the value "Power" received and accepted. Depending on the type of Inverter, the according Default Setpoint is applied (either Default AC Power or Default DC Power), as long as there was no write on either Charge/Discharge AC Power Setpoint or Charge/Discharge DC Power Setpoint with an activated setpoint that is accepted by the Inverter. Applied rules: [COB-914/1], [COB-917].

19: Auto Uncontrolled State --> PCC Control State: Heartbeat and a following write on the Active Control Mode with the value "PCC" received and accepted. The according Default Setpoint is applied (Default PCC Power), as long as there was no write on the PCC Power Setpoint with an activated setpoint that is accepted by the Inverter. Applied rules: [COB-914/2], [COB-917].

2.5 Actors

2.5.1 CEM

The Actor CEM controls other appliances by changing setpoints, shifting flexible demands or activating and deactivating functionalities. In this Use Case the CEM changes the Inverter's Battery functionality by switching between control modes "Power" and "PCC" and setting setpoints and constraints for the charge/discharge power values.

2.5.2 Inverter

The Inverter transforms DC power to AC power and vice versa. It may receive the DC power via PV strings or batteries and may pass on the DC power to a battery.

2.6 Scenarios

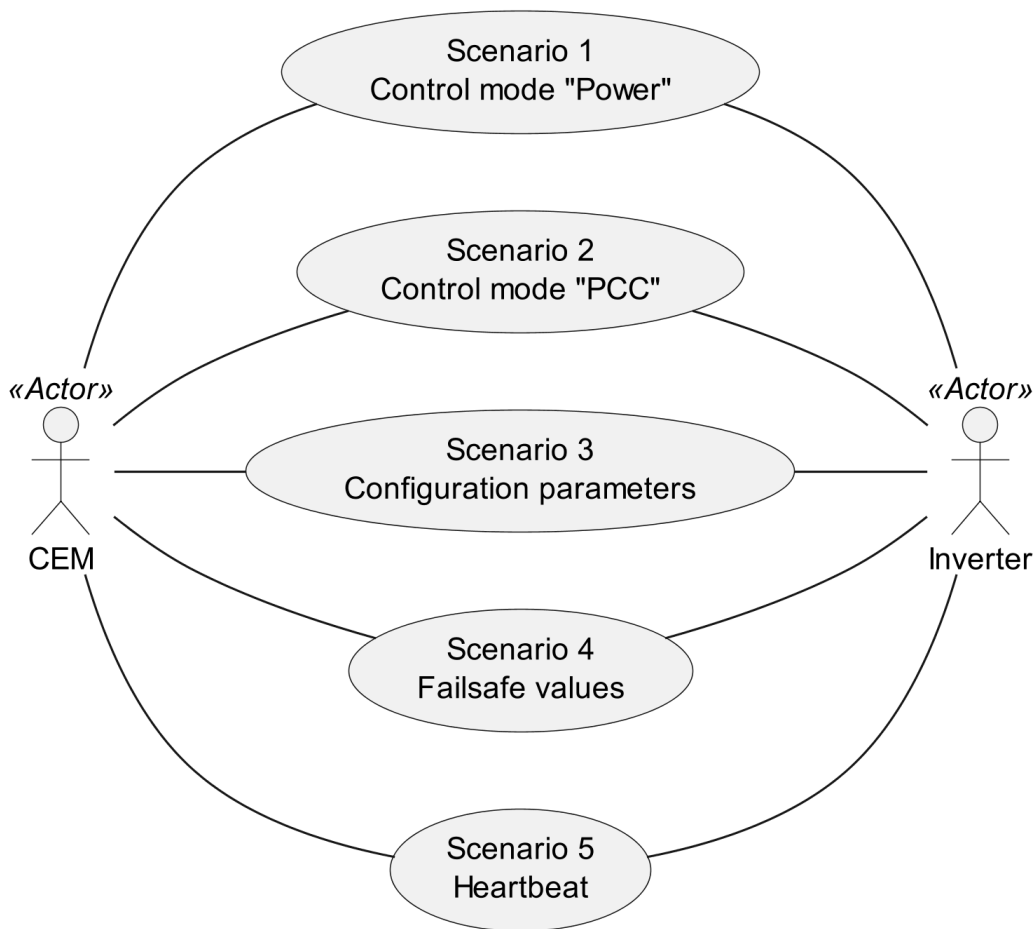


Figure 4: Scenario overview

Scenario number	Scenario name	CEM	Inverter
1	Control mode "Power"	M	M
2	Control mode "PCC"	M	M
3	Configuration parameters	M	M
4	Failsafe values	M	M
5	Heartbeat	M	M

Table 2: Scenario implementation requirements for Actors

2.6.1 Scenario 1 - Control mode "Power"

2.6.1.1 Description

The power the battery or batteries connected to the Inverter consume (charge) or produce (discharge) within "Power Control State" can be set with this Scenario.

In case of a battery Inverter ([COB-970/3]), the setpoint for the AC power of the battery or batteries is set (Charge/Discharge AC Power Setpoint, [COB-011]), as the only power input is AC power.

In case of a hybrid Inverter ([COB-970/2]), the setpoint for the DC power of the battery or batteries is set (Charge/Discharge DC Power Setpoint, [COB-012]), as the power could be gained from the DC input from the PV panel(s), too.

The Inverter SHALL confirm an accepted setpoint (ACK, [COB-002/1]). If the setpoint cannot be applied by the Inverter, the CEM SHALL be informed (NACK, [COB-003/1]). In this case, the old setpoint is not changed and the Inverter will keep up its operation according to the old setpoint.

The CEM MAY activate or deactivate a setpoint ([COB-004/1]). A deactivated setpoint has no effect on the Inverter. In case the setpoint has a duration set, the duration keeps decreasing while the setpoint is deactivated.

The CEM may send a new setpoint at any time.

The Inverter SHALL provide constraints on the setpoints ([COB-980/1]). The Inverter SHALL provide the minimum setpoint value ([COB-980/1/1]) and the maximum setpoint value ([COB-980/1/2]). The Inverter SHOULD provide the setpoint value's step size ([COB-980/1/3]).

A setpoint MAY have a duration that states the time the setpoint is valid for ([COB-005/1]).

The duration will start decreasing immediately after the receipt of the write command, regardless whether the setpoint is activated or deactivated.

As soon as the duration expires (reaches the value "0s"), the Inverter SHALL deactivate the setpoint ([COB-006/1]), and the Inverter MAY remove the duration.

If the Inverter does not receive Heartbeats from the CEM anymore, the setpoint(s) will be deactivated until the CEM sends a Heartbeat again and the CEM performed a proper write command on the Active Control Mode and the setpoint(s) (see section 2.3 and section 2.4 for details).

The Inverter SHALL set each setpoint to "activated" ([COB-009/1]) or "deactivated" ([COB-009/2]) according to the Inverter's state (see section 2.4) ([COB-009]).

Note: The CEM is responsible to consider so-called "race conditions" ([COB-010/1]). I.e. a CEM may just want to adjust a setpoint's value while at the same time the previously set duration expires at the Inverter and the setpoint is deactivated.

In case of a deactivated power setpoint with the Active Control Mode set to "Power", the Inverter SHALL use the value stated in "Default AC Power" ([COB-032]) for pure battery Inverters or "Default DC Power" ([COB-033]) for hybrid Inverters (see [COB-917]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Charge/Discharge AC Power Setpoint	Setpoint for the aggregated AC power the battery or batteries connected to the Inverter SHALL consume (charge) or produce (discharge), if not prevented by technical or regulatory reasons.	M* ¹	[COB-011]* ³
Charge/Discharge DC Power Setpoint	Setpoint for the aggregated DC power the battery or batteries connected to the Inverter SHALL consume (charge) or produce (discharge), if not prevented by technical or regulatory reasons.	M* ²	[COB-012]* ³

Table 3: Scenario 1 - Control mode "Power" - Data point list

*¹: SHALL only be used in case of battery Inverters (see section 2.2.1).

*²: SHALL only be used in case of hybrid Inverters (see section 2.2.1).

*³: For sign convention see section 2.8.1.

2.6.1.2 Conditions

Triggering Event:

The Actor CEM wants to update, activate or deactivate an Inverter's power setpoint.

Pre-condition:

The setpoint for the power of the Actor Inverter's battery or batteries is outdated.

Post-condition:

The Actor Inverter's setpoint for the power of its battery or batteries is updated.

2.6.2 Scenario 2 - Control mode "PCC"

2.6.2.1 Description

In the control mode "PCC", the Inverter monitors the power at the grid connection point (GCP) and adapts its consumption (charging the battery or batteries) or production (discharging the battery or batteries) within "PCC Control State" to match a configured PCC Power Setpoint ([COB-021]).

The Inverter's charge and discharge power may be limited by the following maximum values:

- Max AC Charge Power ([COB-022]), used by battery Inverters
- Max AC Discharge Power ([COB-023]), used by battery Inverters
- Max DC Charge Power ([COB-024]), used by hybrid Inverters
- Max DC Discharge Power ([COB-025]), used by hybrid Inverters

The PCC Power Setpoint is a "soft" setpoint as (e.g.) the following exceptions are permitted:

- Additional energy may be fed into the grid if more power is produced (e.g. by the PV panels) than is consumed in the local grid (household or premises) and all connected batteries are full.

- 723 - Additional energy may be consumed from the grid if more energy is consumed by the local
724 grid (household or premises) than is produced (e.g., by the PV panels) and all connected
725 batteries are empty.
- 726 The Inverter SHALL confirm an accepted setpoint (ACK, [COB-002/1]). If the setpoint cannot be
727 applied by the Inverter, the CEM SHALL be informed (NACK, [COB-003/1]). In this case, the old
728 setpoint is not changed and the Inverter will keep up its operation according to the old setpoint. The
729 same holds valid for the maximum AC/DC values ([COB-022] to [COB-025]) within this Scenario
730 ([COB-020]).
- 731 The CEM MAY activate or deactivate a setpoint ([COB-004/2]). A deactivated setpoint has no effect
732 on the Inverter. In case the setpoint has a duration set, the duration keeps decreasing while the
733 setpoint is deactivated.
- 734 The CEM may send a new setpoint at any time.
- 735 The Inverter SHALL provide constraints on the setpoint ([COB-980/2]). The Inverter SHALL provide
736 the minimum setpoint value ([COB-980/2/1]) and the maximum setpoint value ([COB-980/2/2]). The
737 Inverter SHOULD provide the setpoint value's step size ([COB-980/2/3]).
- 738 A setpoint MAY have a duration that states the time the setpoint is valid for ([COB-005/2]).
- 739 The duration will start decreasing immediately after the receipt of the write command, regardless
740 whether the setpoint is activated or deactivated.
- 741 As soon as the duration expires (reaches the value "0s"), the Inverter SHALL deactivate the setpoint
742 ([COB-006/2]), and the Inverter MAY remove the duration.
- 743 If the Inverter does not receive Heartbeats from the CEM anymore, the setpoint(s) will be
744 deactivated until the CEM sends a Heartbeat again and the CEM performed a proper write command
745 on the Active Control Mode and the setpoint(s) (see section 2.3 and section 2.4 for details).
- 746 The Inverter SHALL set each setpoint to "activated" ([COB-009/1]) or "deactivated" ([COB-009/2])
747 according to the Inverter's state (see section 2.4) ([COB-009]).
- 748 Note: The CEM is responsible to consider so-called "race conditions" ([COB-010/2]). I.e. a CEM may
749 just want to adjust a setpoint's value while at the same time the previously set duration expires at
750 the Inverter and the setpoint is deactivated.
- 751 The maximum (dis)charge AC/DC values are only valid as long as the PCC Power Setpoint has a valid
752 value (setpoint is active and has not been expired) and the control mode "PCC" ([COB-031/2], see
753 section 2.6.3) is active.
- 754 In case of a deactivated PCC Power Setpoint with the control mode set to "PCC", the Inverter SHALL
755 use the value stated in "Default PCC Power" ([COB-034]) (see [COB-917]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
PCC Power Setpoint	Setpoint for the power at the GCP.	M	[COB-021]* ³
Max AC Charge Power	Maximum AC charge power (aggregated) for all connected batteries.	M* ¹	[COB-022]* ³

Max AC Discharge Power	Maximum AC discharge power (aggregated) for all connected batteries.	M ^{*1}	[COB-023] ^{*3}
Max DC Charge Power	Maximum DC charge power (aggregated) for all connected batteries.	M ^{*2}	[COB-024] ^{*3}
Max DC Discharge Power	Maximum DC discharge power (aggregated) for all connected batteries.	M ^{*2}	[COB-025] ^{*3}

Table 4: Scenario 2 - Control mode "PCC" - Data point list

*¹: SHALL only be used in case of battery Inverters (see section 2.2.1).

*²: SHALL only be used in case of hybrid Inverters (see section 2.2.1).

*³: For sign convention see section 2.8.1.

2.6.2.2 Conditions

Triggering Event:

The Actor CEM wants to update, activate or deactivate the Inverter's PCC Power Setpoint.

Pre-condition:

The Inverter's PCC Power Setpoint is outdated.

Post-condition:

The Actor Inverter's PCC Power Setpoint is updated.

2.6.3 Scenario 3 - Configuration parameters

2.6.3.1 Description

Within the "Configuration parameters" Scenario, the current Active Control Mode ([COB-031]) can be changed. Up to this version of this Use Case, the following values are permitted:

- Power
- PCC
- Auto

The Inverter SHALL state whether or not the Active Control Mode may be changed by the CEM ([COB-030/1]) (not permitting the control mode to be changed may be necessary during trickle charging, e.g.). This value may change over time (change of the Active Control Mode possible or not).

The CEM is not permitted to change the Active Control Mode to the value "Auto" ([COB-030/3]).

If a CEM tries to change the Active Control Mode while it is not permitted due to [COB-030/1] or [COB-030/3], the CEM must be aware that the write command will be rejected by the Inverter (see [COB-003/2]).

Additionally, default values for the AC ([COB-032]), DC ([COB-033]) and PCC power ([COB-034]) are provided. These default values SHALL NOT be changeable by the CEM ([COB-030/2]).

785 The Inverter SHALL confirm an accepted Configuration Parameter (ACK, [COB-002/2]). If the
 786 Configuration Parameter cannot be applied or is rejected by the Inverter, the CEM SHALL be
 787 informed (NACK, [COB-003/2]). In this case, the old Configuration Parameter is not changed. Up to
 788 this version of this specification, both rules ([COB-002/2] and [COB-003/2]) hold valid only for [COB-
 789 031] of this Scenario.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Active Control Mode	Current active control mode of the Inverter. Possible values: - Power [COB-031/1] - PCC [COB-031/2] - Auto [COB-031/3]	M	[COB-031]
Default AC Power	Default power value the Inverter applies if it is in control mode "Power" and no valid Charge/Discharge AC power setpoint is set. Typically set to "0" (idle mode).	M* ¹	[COB-032]* ⁴
Default DC Power	Default power value the Inverter applies if it is in control mode "Power" and no valid Charge/Discharge DC power setpoint is set. Typically set to "0" (idle mode).	M* ²	[COB-033]* ⁴
Default PCC Power	Default power value the Inverter applies if it is in control mode "PCC" and no valid PCC power setpoint is set. Typically set to "0" (trying to regulate the GCP to "0").	M* ³	[COB-034]* ⁴

790 *Table 5: Scenario 3 - Configuration parameters - Data point list*

791 *¹: SHALL only be used in case of battery Inverters (see section 2.2.1).

792 *²: SHALL only be used in case of hybrid Inverters (see section 2.2.1).

793 *³: SHALL only be used if control mode "PCC" is supported (see section 2.1.2).

794 *⁴: For sign convention see section 2.8.1.

795

796 **2.6.3.2 Conditions**

797 **Triggering Event:**

798 The Actor CEM wants to change the Active Control Mode of the Actor Inverter or read the default
 799 power values.

800 **Pre-condition:**

801 The Active Control Mode of the Actor Inverter is outdated or the Actor CEM does not know the
 802 default power values.

803 **Post-condition:**

804 The Actor Inverter has an updated Active Control Mode or the Actor CEM does know the default
 805 power values.

806

2.6.4 Scenario 4 - Failsafe values

2.6.4.1 Description

The Failsafe Values within this Scenario are used by the "Failsafe State" (see section 2.4.2) that is used to control the Actor Inverter in case no Heartbeat is received from the Actor CEM. In state "Init" (i.e. after any kind of (re)start of the Inverter) only the Failsafe Setpoints ([COB-041] to [COB-043]) are used. See section 2.3 for details.

The Failsafe Setpoints ([COB-041] to [COB-043]) denote setpoints for the Active Control Modes "Power" and "PCC" in the above-mentioned states. A value SHALL be pre-configured for [COB-041] to [COB-043] (e.g. via a user interface). The values [COB-041] to [COB-043] MAY be changed by the CEM ([COB-040]).

If the Active Control Mode is set to "Auto" while the Inverter is in state "Init" or "Failsafe State", then the Inverter SHALL use [COB-043] as Failsafe Setpoint.

As soon as the Actor CEM changed a value in the Inverter, the respective value SHOULD NOT be configurable via a user interface anymore (or at least a clear indication should be given that changing the value could possibly contradict the overall energy management).

The Failsafe Duration Minimum ([COB-044]) is used to permit the Inverter to leave "Failsafe State" in case the Heartbeat is missing for a longer period of time, which would unnecessarily block the proper functioning of the Inverter (e.g., by a broken CEM). The value SHALL be pre-configured by the Inverter's vendor in the range of 2 hours to 24 hours ([COB-044/1]). The CEM MAY change the value ([COB-044/2]). The Inverter's maximum value for the Failsafe Duration Minimum is defined as the maximum value the Inverter accepts as write command from the CEM. This maximum value SHALL be in the range of the pre-configured value and 24 hours. The CEM SHALL choose a value between 2 hours and 24 hours ([COB-044/3]). The Inverter MAY reject a write command of the CEM on the Failsafe Duration Minimum if the submitted value is greater than the Inverter's maximum value for the Failsafe Duration Minimum ([COB-044/4]). If the Inverter rejects the write command for the aforementioned reason, it SHALL afterwards change the Failsafe Duration Minimum ([COB-044]) to the Inverter's maximum value for the Failsafe Duration Minimum ([COB-044/5]), unless the Failsafe Duration Minimum is already set to the Inverter's maximum value for the Failsafe Duration Minimum.

The Inverter SHALL confirm an accepted Failsafe Value (ACK, [COB-002/3]). If the Failsafe Value cannot be applied by the Inverter, the CEM SHALL be informed (NACK, [COB-003/3]). In this case, the old Failsafe Value is not changed. Both rules ([COB-002/3] and [COB-003/3]) hold valid for all data points in this Scenario ([COB-041] and [COB-044]).

The Inverter SHOULD store changed values of this Scenario persistently.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Failsafe Charge/Discharge AC Power Setpoint	Failsafe setpoint for the Charge/Discharge AC Power Setpoint. If supported, an initial value of "0" SHOULD be configured. The value MAY be changed by the CEM. This setpoint is used in "Init" state or "Failsafe State".	M* ¹	[COB-041]* ⁴

Failsafe Charge/Discharge DC Power Setpoint	Failsafe value for the Charge/Discharge DC Power Setpoint. If supported, an initial value of "0" SHOULD be configured. The value MAY be changed by the CEM. This setpoint is used in "Init" state or "Failsafe State".	M* ²	[COB-042]* ⁴
Failsafe PCC Power Setpoint	Failsafe value for the PCC power setpoint. If supported, an initial value of "0" SHOULD be configured. The value MAY be changed by the CEM. This setpoint is used in "Init" state or "Failsafe State".	M* ³	[COB-043]* ⁴
Failsafe Duration Minimum	Minimum time the Inverter remains in "Failsafe State" unless conditions specified in this Use Case permit leaving the "Failsafe State" (see section 2.3).	M	[COB-044]

Table 6: Scenario 4 - Failsafe values - Data point list

*¹: This value SHALL only be provided if [COB-011] is supported.

*²: This value SHALL only be provided if [COB-012] is supported.

*³: This value SHALL only be provided if [COB-021] is supported.

*⁴: For sign convention see section 2.8.1.

2.6.4.2 Conditions

Triggering Event:

The Actor CEM wants to request the Inverter's Failsafe Values or change the Inverter's Failsafe Values.

Pre-condition:

The Actor CEM does not know the Inverter's Failsafe Values or the Actor Inverter's Failsafe Values are outdated.

Post-condition:

The Actor CEM does know the Inverter's Failsafe Values or the Actor Inverter has new Failsafe Values.

2.6.5 Scenario 5 - Heartbeat

2.6.5.1 Description

The Actors CEM and Inverter periodically send a Heartbeat message to the other Actor. This informs the communication partners CEM and Inverter mutually that the other is working correctly and can be reached. A missing of the Heartbeat from the Actor CEM indicates either a communication problem or technical problems at the CEM. A missing of the Heartbeat from the Actor Inverter indicates either a communication problem or technical problems at the Inverter.

Note: In this Use Case, the behaviour of the CEM in case of a missing Heartbeat of the Inverter is not described, as this is left to the business logic of the CEM, which is outside the scope of this Use Case.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Heartbeat of CEM	The periodically sent message from the CEM. The message SHALL be sent at least every 60 seconds ([COB-007]).	M	[COB-051]
Heartbeat of Inverter	The periodically sent message from the Inverter. The message SHALL be sent at least every 60 seconds ([COB-008]).	M	[COB-052]

Table 7: Scenario 5 - Heartbeat - Data point list

2.6.5.2 Conditions

Triggering Event:

The time is up to send the next Heartbeat.

Pre-condition:

Communication between the CEM and the Inverter is possible.

Post-condition:

The Inverter or the CEM received a Heartbeat.

2.7 Dependencies to other Use Cases

2.7.1 "Limitation of Power Consumption"

The Use Case "Limitation of Power Consumption" (LPC) enables an Energy Guard to limit the power consumption of a Controllable System capable of consuming energy.

If both, "Limitation of Power Consumption" and this Use Case are supported, the limit of "Limitation of Power Consumption" SHALL have a higher priority as the control modes of this Use Case ([COB-952]). This means that eventually the setpoints from this Use Case cannot be hold since the limit of "Limitation of Power Consumption" would not be kept otherwise.

The Actor CEM MAY support the Use Case "Limitation of Power Consumption".

The Actor CEM of this Use Case acts as Actor Energy Guard within the Use Case "Limitation of Power Consumption".

The Actor Inverter MAY support the Use Case "Limitation of Power Consumption".

The Actor Inverter of this Use Case acts as Actor Controllable System within the Use Case "Limitation of Power Consumption".

The following data points of this Use Case SHALL be the same as for the Use Case "Limitation of Power Consumption":

- Failsafe Duration Minimum ([COB-044] and [LPC-022])

This means that if both Use Cases are supported on the same appliance, the data point is provided only once and is used for both Use Case instances.

895 If both Use Cases are supported on the same Inverter, only one CEM/Energy Guard is permitted to
896 control the according functionality.

897 The LPC's state machine is independent from the one of this Use Case (and vice versa). This means
898 that if the Actor Controllable System leaves, e.g., the Failsafe State within LPC, the Actor Inverter
899 does not automatically leave the COB's Failsafe State, although both Use Cases are implemented on
900 the same appliance.

901

902 **2.7.2 "Limitation of Power Production"**

903 The Use Case "Limitation of Power Production" (LPP) enables the Energy Guard to limit the power
904 production of a Controllable System capable of producing energy.

905 If both, "Limitation of Power Production" and this Use Case are supported, the limit of "Limitation of
906 Power Production" SHALL have a higher priority as the control modes of this Use Case ([COB-953]).
907 This means that eventually the setpoints from this Use Case cannot be hold since the limit of
908 "Limitation of Power Production" would not be kept otherwise.

909 The Actor CEM MAY support the Use Case "Limitation of Power Production".

910 The Actor CEM of this Use Case acts as Actor Energy Guard within the Use Case "Limitation of Power
911 Production".

912 The Actor Inverter MAY support the Use Case "Limitation of Power Production".

913 The Actor Inverter of this Use Case acts as Actor Controllable System within the Use Case "Limitation
914 of Power Production".

915 The following data points of this Use Case SHALL be the same as for the Use Case "Limitation of
916 Power Production":

917 - Failsafe Duration Minimum ([COB-044] and [LPP-022])

918 This means that if both Use Cases are supported on the same appliance, the data point is provided
919 only once and is used for both Use Case instances.

920 If both Use Cases are supported on the same Inverter, only one CEM/Energy Guard is permitted to
921 control the according functionality.

922 The LPP's state machine is independent from the one of this Use Case (and vice versa). This means
923 that if the Actor Controllable System leaves, e.g., the Failsafe State within LPP, the Actor Inverter
924 does not automatically leave the COB's Failsafe State, although both Use Cases are implemented on
925 the same appliance.

926

927 **2.7.3 "Monitoring of Inverter"**

928 The Use Case "Monitoring of Inverter" (MOI) describes the information provided by the Actor
929 Inverter.

930 The Actor CEM SHOULD support the Use Case "Monitoring of Inverter".

931 The Actor Inverter SHOULD support the Use Case "Monitoring of Inverter".

932 The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring
933 of Inverter".

934 The Actor Inverter of this Use Case acts as Actor Inverter within the Use Case "Monitoring of
935 Inverter".

936

937 **2.7.4 "Monitoring of Battery"**

938 In case of battery inverters or hybrid inverters, the Use Case "Monitoring of Battery" (MOB) enables
939 to monitor additional parameters of the battery (or batteries).

940 The Actor CEM MAY support the Use Case "Monitoring of Battery".

941 The Actor Inverter SHOULD support the Use Case "Monitoring of Battery", providing the
942 corresponding Actor Battery as so-called "child". This means that the Actor Battery is located behind
943 the Actor Inverter in a hierarchical "device tree" (see section 2.2.1).

944 The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring
945 of Battery".

946 The Actor Inverter of this Use Case acts as parent Actor of the Actor Battery within the Use Case
947 "Monitoring of Battery".

948

949 **2.7.5 "Monitoring of PV String"**

950 In case of hybrid inverters, the Use Case "Monitoring of PV String" (MPS) enables to monitor
951 additional parameters of the PV String(s).

952 The Actor CEM MAY support the Use Case "Monitoring of PV String".

953 The Actor Inverter SHOULD support the Use Case "Monitoring of PV String", providing the
954 corresponding Actor PV String as so-called "child". This means that the Actor PV String is located
955 behind the Actor Inverter in a hierarchical "device tree" (see section 2.2). This only holds valid for
956 hybrid inverters, as pure battery inverters don't have PV strings attached.

957 The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring
958 of PV String".

959 The Actor Inverter of this Use Case acts as parent Actor of the Actor PV String within the Use Case
960 "Monitoring of PV String".

961

962 **2.7.6 Monitoring of Power Consumption**

963 The Use Case "Monitoring of Power Consumption" (MPC) delivers basic measurement values like the
964 current power, phase specific current, etc. of the Inverter.

965 The Actor CEM SHOULD support the Use Case "Monitoring of Power Consumption".

966 The Actor Inverter SHALL support the Use Case "Monitoring of Power Consumption".

967 The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring
968 of Power Consumption".

969 The Actor Inverter of this Use Case acts as Actor Monitored Unit within the Use Case "Monitoring of
970 Power Consumption".

971

972 **2.7.7 "Monitoring of Grid Connection Point"**

973 The Use Case "Monitoring of Grid Connection Point" (MGCP) enables the CEM and the Inverter to
974 monitor the power consumption or production at the Grid Connection Point. This is especially
975 important for the Active Control Mode "PCC".

976 The Actor CEM MAY support the Use Case "Monitoring of Grid Connection Point" in order to track
977 whether the Actor Inverter fulfils the PCC Power Setpoint.

978 The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring
979 of Grid Connection Point".

980 The Actor Inverter SHALL support the Use Case "Monitoring of Grid Connection Point" if Scenario 2 is
981 supported to be able to realize the Active Control Mode "PCC". If Scenario 2 is not supported by the
982 Inverter, supporting the Use Case "Monitoring of Grid Connection Point" is not necessary.

983 The Actor Inverter of this Use Case acts as Actor Monitoring Appliance within the Use Case
984 "Monitoring of Grid Connection Point".

985

986 **2.8 Further information and rules**

987 **2.8.1 Sign conventions**

988 The following values of this Use Case follow the passive sign convention (positive values indicate
989 consumption, negative values indicate production) ([COB-001/1]):

- 990 - Charge/Discharge AC Power Setpoint ([COB-011])
- 991 - Charge/Discharge DC Power Setpoint ([COB-012])
- 992 - PCC Power Setpoint ([COB-021])
- 993 - Default AC Power ([COB-032])
- 994 - Default DC Power ([COB-033])
- 995 - Default PCC Power ([COB-034])
- 996 - Failsafe Charge/Discharge AC Power Setpoint ([COB-041])
- 997 - Failsafe Charge/Discharge DC Power Setpoint ([COB-042])
- 998 - Failsafe PCC Power Setpoint ([COB-043])

999 The following values are always positive. Their name indicates whether they are used for
1000 consumption or production direction ([COB-001/2]):

- 1001 - Max AC Charge Power ([COB-022]): consumption direction
- 1002 - Max AC Discharge Power ([COB-023]): production direction

- 1003 - Max DC Charge Power ([COB-024]): consumption direction
- 1004 - Max DC Discharge Power ([COB-025]): production direction

1005 The Inverter's constraints on the Charge/Discharge AC Power Setpoint and Charge/Discharge DC
1006 Power Setpoint ([COB-980/1]) as well as the Inverter's constraints on the PCC Power Setpoint ([COB-
1007 980/2]) follow the passive sign convention, too ([COB-001/3]).

1008

1009 **2.9 Assumptions and Prerequisites**

1010 None.

1011

1012 3 Technical SPINE solution

1013 3.1 General rules and information

1014 3.1.1 Underlying technology documents

1015 This technical solution relies on the SPINE Resources Specification version 1.2.0
1016 [ResourceSpecification].

1017 For interoperable connectivity this technical solution relies on:

- 1018 - SPINE Protocol Specification version 1.2.0 [ProtocolSpecification] as application protocol.
- 1019 - SHIP Specification version 1.0.1 [SHIP] as transport protocol.

1020

1021 3.1.2 Use Case discovery rules

1022 Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- 1023 - The string content for the Element "nodeManagementUseCaseData. useCaseInformation.
1024 useCaseSupport. useCaseName" within the Use Case discovery (please refer to
1025 [ProtocolSpecification]) SHALL be "controlOfBattery". The string content SHALL only be
1026 defined by this Use Case (regardless of the Use Case version).
- 1027 - The string content of the Element "nodeManagementUseCaseData. useCaseInformation.
1028 actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to
1029 the according value stated within the corresponding Actor's section.
- 1030 - An Actor A that is implemented to support this Use Case specification SHALL set the Element
1031 "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion"
1032 within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- 1033 - If an Actor A supports multiple versions of this Use Case with the same major version
1034 number, only the highest one SHOULD be set within the Use Case discovery.
- 1035 - If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple
1036 versions of this Use Case with the same major version number as supported by Actor A, the
1037 Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- 1038 - If an Actor A supports multiple versions of this Use Case with different major version
1039 numbers, for each major version number only the highest version number SHOULD be set
1040 within the Use Case discovery.
- 1041 - If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions
1042 with a major version number not implemented by Actor A, it still might be possible to run the
1043 Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B
1044 as a valid partner for this Use Case.

1045

1046 3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

1047 3.1.3.1 General presence indication definitions

1048 Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
--------------	---------	------------------------------

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 8: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

1084 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 1085 - M, O
- 1086 - M \W, O \W

1087 The description (rules) of the Element details in such a case the presence requirement. A typical
1088 example is a list-based Element where the "first" (or "last") Element has a different presence
1089 requirement than the other list Elements.

1090

1091 **3.1.3.3 Presence indications for "Content of Function..." tables**

1092 This section only defines rules for the server side.

1093 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
1094 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
1095 Element, because the Element is set only by the client.

1096 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 1097 - "R" means that the data SHOULD be provided by the server.
- 1098 - "O" means that the data MAY be provided by the server.
- 1099 - "F" means that the data SHALL NOT be provided by the server.

1100 The following applies for writeable data that is exchanged in a "write" operation:

- 1101 - "R" means that the data SHOULD be supported. In other words: If the client writes the
1102 Element, the server SHOULD accept those messages and the contained Elements.
- 1103 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
1104 the server MAY accept those messages and the contained Elements.

1105 The following applies for Elements that are not listed in the Actor section:

- 1106 - In case of a received "read" request: The according Element MAY be set in the reply.
- 1107 - In case of a received "write" operation: The server MAY ignore the Element.
- 1108 - In case of a "notify" operation to be created: The server MAY set the Element.

1109 Note: The server will only accept write operations if the result fulfils the server Function
1110 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
1111 result in an error message.

1112 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
1113 only has to be supported during a certain event and hence does not need to be present at all times. If
1114 the event is not active the Element may be omitted or another value may be set. In most cases a
1115 High-Level requirement reference for the event is given in the rules column.

1116 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 1117 - M, O
- 1118 - M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
No cardinality indication.
2. X (a..b)
This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..
This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 9: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3

1145 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for
 1146 Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

1147 2: M \W

1148 denotes that this data set is mandatory for Scenario 2.

1149 The field

1150 xSlot. (1..)

1151 expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur
 1152 more than one time.

1153 For the expression "<*#{1..}>" of Element "xId" please see section 3.1.3.6.

1154 The remaining fields do not have an explicit cardinality indication.

1155 Note: Cardinality expressions are also used within placeholder expressions as defined in section
 1156 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted
 1157 Elements or list items as they explicitly define how many different values for a given Identifier are
 1158 required or permitted for a given Scenario.

1159

1160 **3.1.3.5 Writability and changeability indication**

1161 In the same column where the presence indications are denoted, a mark is used to distinguish
 1162 between writeable, changeable or readable Elements:

- 1163 - Elements that are marked with "\W" are written by a client and SHALL be writeable at the
 1164 server according to their presence indications. The client is not obliged to read the according
 1165 data. Received notifications do not need to be evaluated.
- 1166 - Elements that are marked with "\C" are changed by a client and SHALL be changeable at the
 1167 server according to their presence indications. The client is not obliged to read the according
 1168 data. Received notifications do not need to be evaluated.
- 1169 - Elements that are marked with "\RW" are read and written by a client and SHALL be
 1170 writeable and provided by the server according to their presence indications. Received
 1171 notifications SHALL be evaluated according to their presence indications.
- 1172 - Elements that are marked with "\RC" are read and changed by a client and SHALL be
 1173 changeable and provided by the server according to their presence indications. Received
 1174 notifications SHALL be evaluated according to their presence indications.
- 1175 - Elements that are not marked are only read by a client and SHALL be provided by the server
 1176 according to their presence indications. Received notifications SHALL be evaluated according
 1177 to their presence indications.

1178 "Writeable" means that the Element and its value may be written by a client. This includes the
 1179 possibility to modify (if the Element is already present), create (if the Element is not present yet), and
 1180 delete the Element. The server SHALL adjust its Function according to the received "write" operation
 1181 (unless the server cannot accept the "write" operation according to section 3.1.3.3).

1182 "Changeable" means that the Element's value may be changed by a client. If the Element is not
1183 present at the resource before, it probably **cannot** be created by the client via the "write" operation.
1184 In this case the server MAY decline such a message.

1185 Note: "\W" includes "\C" already.

1186 Note: Depending on the resource a client might need to request a proper binding before the server
1187 accepts a "write" operation.

1188

1189 **3.1.3.6 "Value" placeholders**

1190 *3.1.3.6.1 Introduction*

1191 Specializations may use placeholders to model relations between different Elements or even list
1192 items of different Functions. The main purpose is to declare which Identifier values relate to each
1193 other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder
1194 like "<x1>" can be used in "Value" columns to express the intended relations.

1195 There are two styles to identify a placeholder that can be referenced:

- 1196 1. <xM>
- 1197 2. <xM#S>

1198 where

- 1199 1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
- 1200 2. "M" is a (major) number like "1", "2", "15", etc.
- 1201 3. "S" is a sub-number like "1", "7", "10", etc.

1202 Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>",
1203 "<m22#3>". For a given placeholder, only one of the styles can be used.

1204 In addition, there are also styles for placeholders that do not need to be referenced:

- 1205 1. <*>
- 1206 2. <*#S>

1207 The second style is only used with so-called cardinality expressions.

1208

1209 *3.1.3.6.2 Uniqueness of placeholders*

1210 A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case
1211 specification for a given set of its parent Identifier values. This shall be explained in a brief example:

1212 We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with
1213 placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of
1214 pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2.
1215 But it also means that "<s1>" represents the same value in all list items with pId=1.

1216 Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g.
 1217 In this case, the uniqueness rule applies for "<p5>" likewise.

1218 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

1219

1220 3.1.3.6.3 Placeholder expressions with cardinalities

1221 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 1222 purpose, which make use of cardinality expressions. The general notation is as follows:

1223 1. <xM#(a..b)>

1224 This is equivalent to this explicit definition:

1225 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

1226 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 1227 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 1228 respective Identifier.

1229 Additionally, the following notations may occur:

1230 2. <xM#(a..)>

1231 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

1232 3. <xM#(..b)>

1233 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

1234 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 1235 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 1236 "b" distinct values.

1237 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

1238 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 1239 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 1240 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 1241 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 1242 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 1243 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

1244

1245 3.1.3.6.4 References to placeholders and relations

1246 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 1247 to a particular placeholder:

1248 1. e(-><xM>)

1249 2. e(-><xM#S>)

1250 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

1251 Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with
 1252 placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using
 1253 this expression:

1254 "a"(-><m2#1>)

1255 "b"(-><m2#2>)

1256 "c"(-><m2#3>)

1257 This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to
 1258 be used with <m2#2> and "c" with <m2#3>.

1259 Similarly, a placeholder "yN" can refer to a particular placeholder:

1260 3. <yN->xM>

1261 4. <yN->xM#S>

1262 5. <yN#T->xM>

1263 6. <yN#T->xM#S>

1264 where "T" is a sub-number of "yN".

1265 It is also feasible to associate placeholders with cardinalities:

1266 7. <yN#(a..b)->xM#(a..b)>

1267 denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

1268 Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

1269 In some cases, there is a need to express that multiple list items with similar values are feasible or
 1270 required, but only particular combinations of these different data are permitted. The following
 1271 example shows that several "fData" list items with different "phase" value are required, but that
 1272 these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase"
 1273 value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a
 1274 table:

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

1275 Table 10: Content of an example Specialization

1276 Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or
1277 {<z3#1>, <z3#4>, <z3#5>}.
1278

1279 **3.1.3.7 Rules for content of "Value" column**

1280 For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element
1281 to one or more particular values. This means that Elements with values deviating from the restriction
1282 (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered
1283 as if the Element is not set. If more than one particular value is permitted for an Element, the values
1284 are in a single line each.

1285 If a presence indication is set for the value (in an additional column before the value), the following
1286 rules SHALL be applied:

- 1287 - "M" means that the value SHALL be supported. This means the value needs to be set at a
1288 certain point in time (depending on the value rules) or for a certain Element within a list
1289 entry.
- 1290 - "R" means that the value SHOULD be supported.
- 1291 - "O" means that the value MAY be supported.

1292 If all possible values of a given mandatory Element are optional or recommended and this Element is
1293 used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values
1294 of a given optional or recommended Element are optional or recommended, this Element MAY
1295 contain also other values, but then this Element SHALL NOT be considered as part of the respective
1296 Scenario.

1297 M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be
1298 supported during a certain event and hence does not need to be present at all times. If the event is
1299 not active another value may be set. In most cases a High-Level requirement reference for the event
1300 is given in the rules column.

1301 If no presence indication is set for the value, the following rules SHALL be applied:

- 1302 - In case of Elements where the server may set or change an Element on its own (see section
1303 3.1.3.5):
 - 1304 ○ within the tables in the "Server data - Resources" sections:
 - 1305 ■ the server SHALL support at least one of the listed values.
 - 1306 ○ within the tables in the "Client data - Specializations" sections:
 - 1307 ■ the client SHALL support all listed values.
- 1308 - In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - 1309 ○ within the tables in the "Server data - Resources" sections:
 - 1310 ■ the server SHALL support all listed values.
 - 1311 ○ within the tables in the "Client data - Specializations" sections:
 - 1312 ■ the client SHALL support at least one of the listed values.

1313 Depending on the Element, different values may be used during runtime. If this is the case, those
1314 rules are described within the value rules.

1315 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 1316 MAY deviate from this, e.g. "(1024)".

1317

1318 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 1319 **Specialization..." tables**

1320 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 1321 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 1322 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 1323 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 1324 background colour. This row contains the following parts:

1325 <Feature Type>. <Function>.[<list entry instance name>.]

1326 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 1327 could be:

1328 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 1329 deviceConfigurationKeyValueDescriptionData.

1330 In the following rows, only the names of the Elements are stated, without the prefix described above.

1331

1332 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 1333 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 1334 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 1335 "Function "measurementListData"). These sections contain one table with all Elements needed for
 1336 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 1337 colour. This row contains the following parts:

1338 <Feature Type>. <Function>.[<list entry instance name>.]

1339 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 1340 could be:

1341 Measurement. measurementDescriptionListData. measurementDescriptionData.

1342 In the following rows, only the names of the Elements are stated, without the prefix described above.

1343

1344 For both kinds of tables, the following applies:

1345 - Parent Elements are marked with a dot at the end of the name:

1346 <parent Element>.

1347 E.g.:

1348 value.

1349 - If there are sub-Elements, they are described in own rows with the name of the parent
 1350 Element as prefix, separated by a dot and a blank space:

1351 <parent Element>. <sub-Element>

1352 E.g.:

1353 value. number

1354

1355 3.1.4 Rules for "Feature Types and Functions..." tables

1356 3.1.4.1 Presence indications for "Feature Types and Functions..." tables

1357 The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

1358 Table 11: Presence indication of Feature Types and Functions support

1359 If at least one Function of a Feature has the presence indication "M", it is mandatory to support the
1360 Feature.

1361

1362 3.1.4.2 Rules for "Possible operations" column

1363 Within the "Feature Types and Functions..." tables the column "Possible operations" state whether
1364 the Function is read- or writeable (as defined in the detailed discovery mechanism, see
1365 [ProtocolSpecification]).

1366 If the "partial" concept (also called "restricted function exchange") SHALL be supported, the
1367 following notation is used (separated for read and write access):

1368 read (M). partial (M)

1369 write (M). partial (M)

1370 If the "partial" concept SHOULD be supported, the following notation is used:

1371 read (M). partial (R)

1372 write (M). partial (R)

1373 If the "partial" concept MAY be supported, the following notation is used:

1374 read (M). partial (O)

1375 write (M). partial (O)

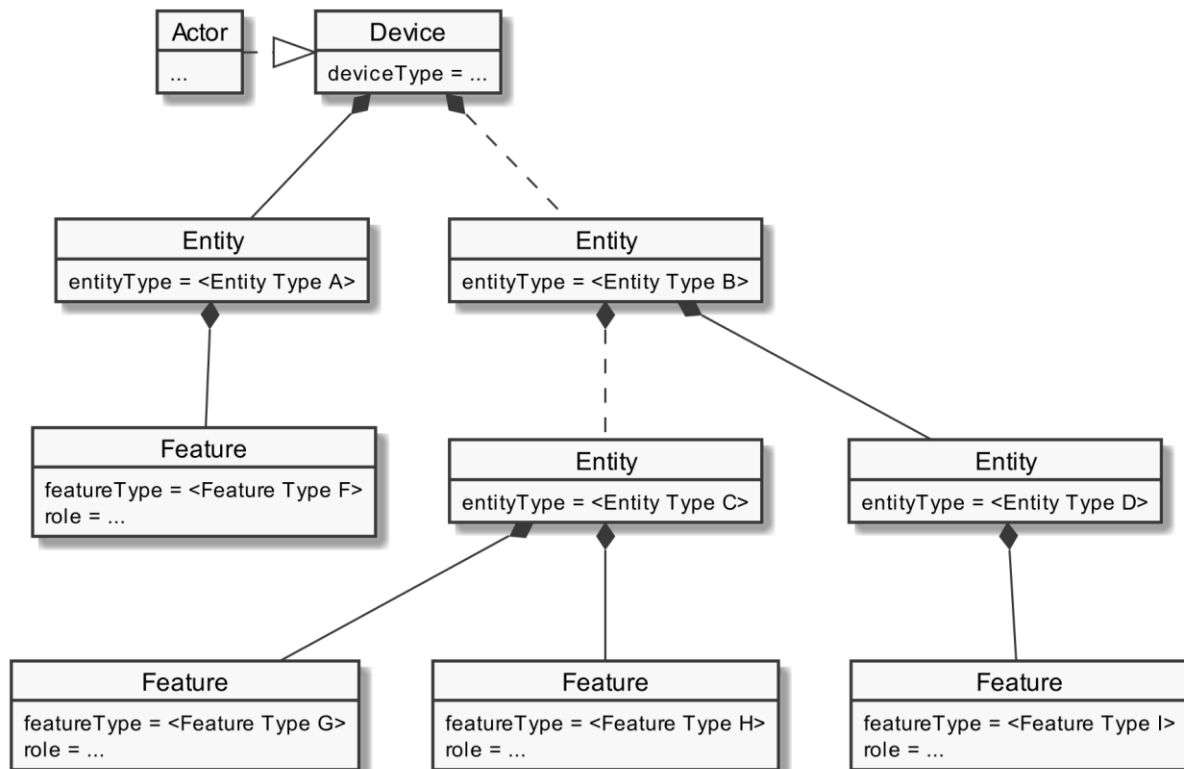
1376 The server can decide whether a notification is submitted complete or partial (as described in
1377 [ProtocolSpecification]) if not defined differently within this Use Case Specification.

1378

1379 3.1.5 "Actor ... overview" diagram rules

1380 Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of
1381 this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in
1382 Table 2. The Actor MAY have more functionality implemented than needed for this Use Case.

1383 For the following Actor overview example, a brief description of the graphical symbols will be
 1384 described.



1385
 1386 *Figure 5: Actor overview example*

1387 The solid lines in the figure represent an immediate parent-childhood relation: The Entity with
 1388 "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the
 1389 Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

1390 The dashed lines in the figure express that there MAY be additional Entities between the shown
 1391 Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity
 1392 with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between
 1393 the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

1394

1395 3.1.6 Specializations

1396 Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset
 1397 necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data
 1398 from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a
 1399 Specialization defines a dataset that may encompass multiple related Functions from one or more
 1400 different Features.

1401 As different Use Cases sometimes share similar requirements, Specializations are also important
 1402 from a re-usability perspective. This approach is used to improve consistency across Use Cases and
 1403 avoid multiple variances of basically the same dataset. This is especially important in the case when
 1404 an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two
 1405 different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula:
$$\text{number} * 10^{\text{scale}}$$

3.1.8.2 Rules regarding the usage of time-related information

Timestamps used within this Use Case SHALL be presented as absolute times. This means relative times SHALL NOT be used for timestamps.

1442 Durations used within this Use Case SHALL be presented as relative times. The same holds for the
 1443 "endTime" Element used for the validity duration ([COB-005/1]) of the Charge/Discharge AC Power
 1444 Setpoint ([COB-011]) or the Charge/Discharge DC Power Setpoint ([COB-012]) and the validity
 1445 duration ([COB-005/2]) of the PCC Power Setpoint ([COB-021]).

1446 Note: This means the Actors need to have a common understanding of absolute times. It is strongly
 1447 recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal
 1448 Time" (UTC).

1449

1450 3.1.8.3 Applied sign convention

1451 For the applied sign convention, please see section 2.8.1.

1452

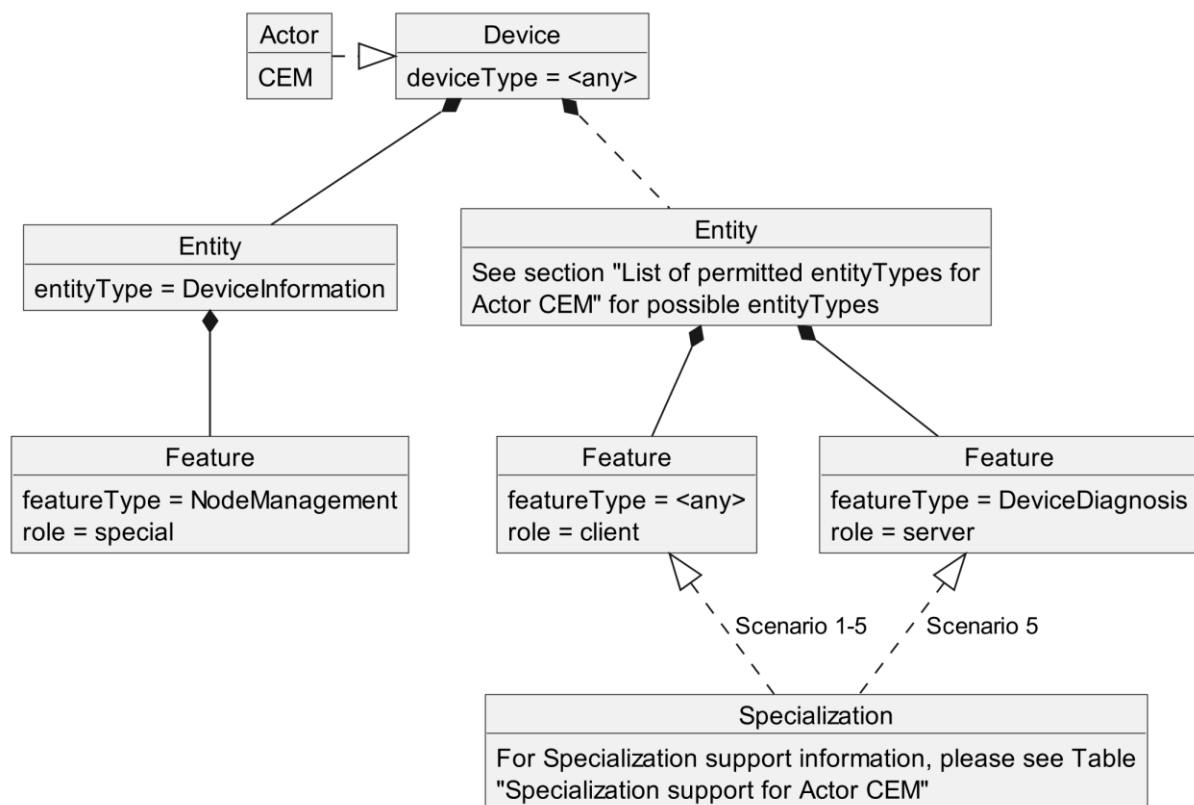
1453 3.2 Actors

1454 3.2.1 CEM

1455 3.2.1.1 Resource hierarchy

1456 If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "CEM" in the
 1457 Element "nodeManagementUseCaseData. useCaseInformation. actor".

1458 The following diagram provides an overview of the Actor CEM's resource hierarchy.



1459

1460 Figure 6: Actor "CEM" overview

1461 The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See
 1462 the "Specializations" section for more information regarding the Specializations given in the diagram
 1463 above.

1464 Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by
 1465 the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for
 1466 completeness but are not directly linked to the Use Case.

1467 The Use Case specific data follows behind one of the entityTypes listed in section 3.2.1.1.1. The
 1468 Specializations represent the Scenario specific data that must be supported for each Scenario and are
 1469 realized through the corresponding featureTypes.

1470 If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this
 1471 data. This means that the Actor accesses the data set described by the Specialization at a
 1472 corresponding server Feature. Further details are described in the sub-section "Client data -
 1473 Specializations".

1474 If a Specialization is connected to a Feature with the role "server", the Actor has the server role for
 1475 this data. This means that the Actor must provide the corresponding data set of the Specialization as
 1476 part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
DeviceDiagnosisHeartbeat_Timeout60Seconds	5	DeviceDiagnosis	Table 14
Specialization name	Scenario	Described in table	
Setpoint_BatteryAcPower	1	Table 15	
Setpoint_BatteryDcPower	1	Table 16	
Setpoint_PccPower	2	Table 17	
DeviceConfiguration_MaxAcChargePower	2	Table 18	
DeviceConfiguration_MaxAcDischargePower	2	Table 19	
DeviceConfiguration_MaxDcChargePower	2	Table 20	
DeviceConfiguration_MaxDcDischargePower	2	Table 21	
DeviceConfiguration_BatteryActiveControlMode	3	Table 22	
DeviceConfiguration_DefaultAcPower	3	Table 23	
DeviceConfiguration_DefaultDcPower	3	Table 24	
DeviceConfiguration_DefaultPccPower	3	Table 25	
DeviceConfiguration_FailsafeAcPowerSetpoint	4	Table 26	
DeviceConfiguration_FailsafeDcPowerSetpoint	4	Table 27	
DeviceConfiguration_FailsafePccPowerSetpoint	4	Table 28	
DeviceConfiguration_FailsafeDurationMinimum	4	Table 29	
DeviceDiagnosisHeartbeat_Timeout60Seconds	5	Table 30	

1477 *Table 12: Specialization support for Actor CEM*

1478

1479 3.2.1.1.1 List of permitted entityTypes for Actor CEM

1480 In this version of this Use Case, only the following entityTypes are permitted for the Actor CEM:

1481 - CEM

1482 This may change at a later version and be extended to further entityTypes.

1483

1484 **3.2.1.2 Server data - Resources**1485 **3.2.1.2.1 Overview**

1486 Behind one of the entityTypes listed in section 3.2.1.1.1, the Actor CEM SHALL offer the Feature
 1487 Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
DeviceDiagnosis	5: M	deviceDiagnosisHeartbeatData	read (M)

1488 *Table 13: Feature Types and Functions used within this Use Case by the Actor CEM*

1489 For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature
 1490 with the Feature Type in the Entity.

1491 Note: As a consequence of the previous rule, an implementation may need to have Feature data
 1492 from different Scenarios/Specializations or even Use Cases in a given Feature.

1493 The Scenario number shows in which Scenarios the CEM acts as server and which Feature Types and
 1494 Functions are relevant in each Scenario.

1495 A detailed definition of the Elements and values that shall be supported in each Function is given in
 1496 the following sub-sections.

1497 Note: If in the table above "partial" read is not mentioned or is only optional, it still might be
 1498 mandatory to support partial notifications. The details of "partial" support are described within the
 1499 Scenario sections.

1500 Note: The presence indications stated above are meant relative to the ones of the according Scenario
 1501 stated in Table 2. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the
 1502 Feature Type need only be supported if the Scenario is supported, too.

1503 Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be
 1504 implemented in the used Entities.

1505

1506 **3.2.1.2.2 Feature Type "DeviceDiagnosis"**1507 **3.2.1.2.2.1 Function "deviceDiagnosisHeartbeatData"**

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
5: M	timestamp		SHALL hold the time of creation.
5: M	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The

			deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
5: M	heartbeatTimeout	≤60 seconds* ¹ , * ² [COB-007]	[COB-051] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 14: Content of Function "deviceDiagnosisHeartbeatData" at Actor CEM

*¹: Could be e.g. "PT60S" or "PT1M"

*²: For the COB Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "Setpoint"

3.2.1.3.1.1 Specialization "Setpoint_BatteryAcPower"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: R	measurementId	<m1#(1..1)>	SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. If the Use Case "Monitoring of Power Consumption" (MPC) is supported by the Actor Inverter, the same measurementId SHALL be used for <m1#1> of this Use Case and <m1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <m1#1> (in the Use Case MPC) is set to "18", <m1#1> (in this Use Case) SHALL be set to "18", too.

1: M	setpointType	"valueAbsolute"	
1: M	unit	"W"	
1: M	scopeType	"batteryAcPower"	
1: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/1]		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointRangeMin.		[COB-001/3], [COB-980/1/1] The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]
1: M	setpointRangeMin. number		SHALL be used.
1: M	setpointRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	setpointRangeMax.		[COB-001/3], [COB-980/1/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
1: M	setpointRangeMax. number		SHALL be used.
1: M	setpointRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	setpointStepSize.		[COB-001/3], [COB-980/1/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
1: M	setpointStepSize. number		SHALL be used.
1: M	setpointStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	Setpoint. setpointListData. setpointData.		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[COB-001/1], [COB-011], [COB-960], [COB-980/1] [Scaled number rules]
1: M \C	value. number		SHALL be used.
1: M \C	value. scale		SHALL be used.
1: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
1: M \C	isSetpointActive	"true" "false"	[COB-004/1], [COB-006/1], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true",

			the timePeriod and value Elements SHALL be applied.
1: M, O \W	timePeriod.		
1: M \RW, O \RW	timePeriod. endTime		[COB-005/1]

Table 15: Content of Specialization "Setpoint_BatteryAcPower" at Actor CEM

3.2.1.3.1.2 Specialization "Setpoint_BatteryDcPower"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
1: M	setpointId	<st2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointType	"valueAbsolute"	
1: M	unit	"W"	
1: M	scopeType	"batteryDcPower"	
1: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/1]		
1: M	setpointId	<st2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointRangeMin.		[COB-001/3], [COB-980/1/1] The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]
1: M	setpointRangeMin. number		SHALL be used.
1: M	setpointRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	setpointRangeMax.		[COB-001/3], [COB-980/1/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
1: M	setpointRangeMax. number		SHALL be used.
1: M	setpointRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	setpointStepSize.		[COB-001/3], [COB-980/1/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
1: M	setpointStepSize. number		SHALL be used.

1: M	setpointStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	Setpoint. setpointListData. setpointData.		
1: M	setpointId	<st2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[COB-001/1], [COB-012], [COB-960], [COB-980/1] [Scaled number rules]
1: M \C	value. number		SHALL be used.
1: M \C	value. scale		SHALL be used.
1: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
1: M \C	isSetpointActive	"true" "false"	[COB-004/1], [COB-006/1], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
1: M, O \W	timePeriod.		
1: M \RW, O \RW	timePeriod. endTime		[COB-005/1]

Table 16: Content of Specialization "Setpoint_BatteryDcPower" at Actor CEM

3.2.1.3.1.3 Specialization "Setpoint_PccPower"

Scenario [...]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
2: M	setpointId	<st3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	setpointType	"valueAbsolute"	
2: M	unit	"W"	
2: M	scopeType	"pccPower"	
2: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/2]		
2: M	setpointId	<st3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	setpointRangeMin.		[COB-001/3], [COB-980/2/1] The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]

2: M	setpointRangeMin. number		SHALL be used.
2: M	setpointRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	setpointRangeMax.		[COB-001/3], [COB-980/2/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
2: M	setpointRangeMax. number		SHALL be used.
2: M	setpointRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	setpointStepSize.		[COB-001/3], [COB-980/2/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
2: M	setpointStepSize. number		SHALL be used.
2: M	setpointStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Setpoint. setpointListData. setpointData.		
2: M	setpointId	<st3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/1], [COB-021], [COB-960], [COB-980/2] [Scaled number rules]
2: M \C	value. number		SHALL be used.
2: M \C	value. scale		SHALL be used.
2: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
2: M \C	isSetpointActive	"true" "false"	[COB-004/2], [COB-006/2], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
2: M, O \W	timePeriod.		
2: M \RW, O \RW	timePeriod. endTime		[COB-005/2]

Table 17: Content of Specialization "Setpoint_PccPower" at Actor CEM

1525 3.2.1.3.2 Topic "DeviceConfiguration"

1526 3.2.1.3.2.1 Specialization "DeviceConfiguration_MaxAcChargePower"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxAcChargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-022] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

1527 Table 18: Content of Specialization "DeviceConfiguration_MaxAcChargePower" at Actor CEM

1528

1529 3.2.1.3.2.2 Specialization "DeviceConfiguration_MaxAcDischargePower"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxAcDischargePower"	
2: M	valueType	"scaledNumber"	

2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-023] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

Table 19: Content of Specialization "DeviceConfiguration_MaxAcDischargePower" at Actor CEM

3.2.1.3.2.3 Specialization "DeviceConfiguration_MaxDcChargePower"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxDcChargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-024] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]

2: M \\C	value. scaledNumber. number		SHALL be used.
2: M \\C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

Table 20: Content of Specialization "DeviceConfiguration_MaxDcChargePower" at Actor CEM

3.2.1.3.2.4 Specialization "DeviceConfiguration_MaxDcDischargePower"

Scenario [{...}]: M/R/O [\\W][\\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxDcDischargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-025] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \\C	value. scaledNumber. number		SHALL be used.
2: M \\C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

Table 21: Content of Specialization "DeviceConfiguration_MaxDcDischargePower" at Actor CEM

1538 3.2.1.3.2.5 Specialization "DeviceConfiguration_BatteryActiveControlMode"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"batteryActiveControlMode"	
3: M	valueType	"string"	
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-031] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M \C	value. string	"power"	[COB-031/1]
		"pcc"	[COB-031/2]
		"auto"	[COB-031/3]
3: M	isValueChangeable	"true" "false"	[COB-030/1] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.

1539 Table 22: Content of Specialization "DeviceConfiguration_BatteryActiveControlMode" at Actor CEM

1540

1541 3.2.1.3.2.6 Specialization "DeviceConfiguration_DefaultAcPower"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultAcPower"	
3: M	valueType	"scaledNumber"	
3: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

3: M	value.		[COB-001/1], [COB-032] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2] If set to "false" the server SHALL decline change requests.

Table 23: Content of Specialization "DeviceConfiguration_DefaultAcPower" at Actor CEM

3.2.1.3.2.7 Specialization "DeviceConfiguration_DefaultDcPower"

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultDcPower"	
3: M	valueType	"scaledNumber"	
3: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-001/1], [COB-033] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2]

			If set to "false" the server SHALL decline change requests.
--	--	--	---

Table 24: Content of Specialization "DeviceConfiguration_DefaultDcPower" at Actor CEM

3.2.1.3.2.8 Specialization "DeviceConfiguration_DefaultPccPower"

Scenario [...]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultPccPower"	
3: M	valueType	"scaledNumber"	
3: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-001/1], [COB-034] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2] If set to "false" the server SHALL decline change requests.

Table 25: Content of Specialization "DeviceConfiguration_DefaultPccPower" at Actor CEM

3.2.1.3.2.9 Specialization "DeviceConfiguration_FailsafeAcPowerSetpoint"

Scenario [...]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
-----------------------------------	---------	-------	---

4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k9#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeAcPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k9#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-041] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \C	value. scaledNumber. number		SHALL be used.
4: M \C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

Table 26: Content of Specialization "DeviceConfiguration_FailsafeAcPowerSetpoint" at Actor CEM

3.2.1.3.2.10 Specialization "DeviceConfiguration_FailsafeDcPowerSetpoint"

Scenario [...]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeDcPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-042]

			Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \C	value. scaledNumber. number		SHALL be used.
4: M \C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

1554 Table 27: Content of Specialization "DeviceConfiguration_FailsafeDcPowerSetpoint" at Actor CEM

1555

1556 3.2.1.3.2.11 Specialization "DeviceConfiguration_FailsafePccPowerSetpoint"

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k11#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafePccPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k11#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-043] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \C	value. scaledNumber. number		SHALL be used.
4: M \C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040]

			If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
--	--	--	---

Table 28: Content of Specialization "DeviceConfiguration_FailsafePccPowerSetpoint" at Actor CEM

3.2.1.3.2.12 Specialization "DeviceConfiguration_FailsafeDurationMinimum"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k12#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeDurationMinimum"	
4: M	valueType	"duration"	
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k12#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-044] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M \C	value. duration	[2h..24h]* ¹	
4: M	isValueChangeable	"true"	[COB-044/2] If set to "true" the server SHALL evaluate changes of the element "value" by appropriate clients (e.g. the bound client) according to rules stated in [COB-044/4].

Table 29: Content of Specialization "DeviceConfiguration_FailsafeDurationMinimum" at Actor CEM

*¹: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

1564 3.2.1.3.3 Topic "DeviceDiagnosisHeartbeat"

1565 3.2.1.3.3.1 Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
5: M	timestamp		SHALL hold the time of creation.
5: R ^{*3}	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
5: R ^{*3}	heartbeatTimeout	≤60 seconds ^{*1} , ^{*2} [COB-008]	[COB-052] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

1566 Table 30: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor CEM

1567 ^{*1}: Could be e.g. "PT60S" or "PT1M"

1568 ^{*2}: For the COB Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may
 1569 also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout
 1570 SHALL be ≤60 seconds.

1571 ^{*3}: The CEM is not obliged to verify this Element within the Inverter's Heartbeat message.

1572

1573 3.2.2 Inverter

1574 3.2.2.1 Resource hierarchy

1575 If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "Inverter" in the
 1576 Element "nodeManagementUseCaseData. useCaseInformation. actor".

1577 The following diagram provides an overview of the Actor Inverter's resource hierarchy.

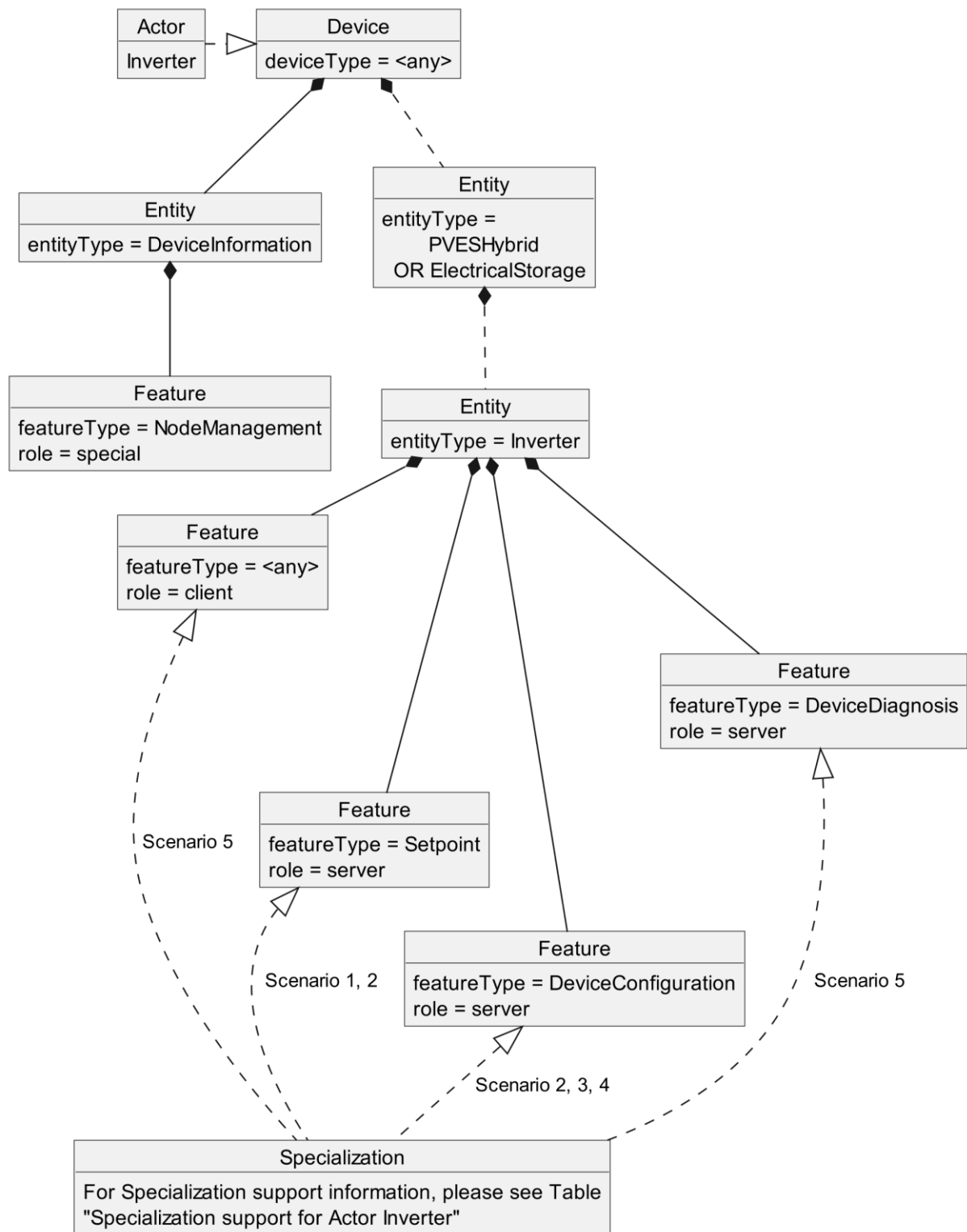


Figure 7: Actor "Inverter" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

1586 The Use Case specific data follows behind the entityType "Inverter" which is a sub-Entity of one of
 1587 the following Entities: "PVESHybrid" ([COB-970/2]) or "ElectricalStorage" ([COB-970/3]). The
 1588 Specializations represent the Scenario specific data that must be supported for each Scenario and are
 1589 realized through the corresponding featureTypes.

1590 If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this
 1591 data. This means that the Actor accesses the data set described by the Specialization at a
 1592 corresponding server Feature. Further details are described in the sub-section "Client data -
 1593 Specializations".

1594 If a Specialization is connected to a Feature with the role "server", the Actor has the server role for
 1595 this data. This means that the Actor must provide the corresponding data set of the Specialization as
 1596 part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
Setpoint_BatteryAcPower	1	Setpoint	Table 33
			Table 34
			Table 35
Setpoint_BatteryDcPower	1	Setpoint	Table 33
			Table 34
			Table 35
Setpoint_PccPower	2	Setpoint	Table 33
			Table 34
			Table 35
DeviceConfiguration_MaxAcChargePower	2	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_MaxAcDischargePower	2	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_MaxDcChargePower	2	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_MaxDcDischargePower	2	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_BatteryActiveControlMode	3	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_DefaultAcPower	3	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_DefaultDcPower	3	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_DefaultPccPower	3	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_FailsafeAcPowerSetpoint	4	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_FailsafeDcPowerSetpoint	4	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_FailsafePccPowerSetpoint	4	DeviceConfiguration	Table 36
			Table 37
DeviceConfiguration_FailsafeDurationMinimum	4	DeviceConfiguration	Table 36
			Table 37
DeviceDiagnosisHeartbeat_Timeout60Seconds	5	DeviceDiagnosis	Table 38
Specialization name	Scenario	Described in table	

DeviceDiagnosisHeartbeat_Timeout60Seconds	5	Table 39
---	---	----------

Table 31: Specialization support for Actor Inverter

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind the entityType "Inverter", the Actor Inverter SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/ O	Function	Possible operations
Setpoint	1: M 2: M	setpointDescriptionListData	read (M). partial (R)
	1: M 2: M	setpointConstraintsListData	read (M). partial (R)
	1: M 2: M	setpointListData	read (M). partial (R) write (M). partial (M)
DeviceConfiguration	2: M 3: M 4: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	2: M 3: M 4: M	deviceConfigurationKeyValueListData	read (M). partial (R) write (M). partial (M)
DeviceDiagnosis	5: M	deviceDiagnosisHeartbeatData	read (M)

Table 32: Feature Types and Functions used within this Use Case by the Actor Inverter

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Inverter acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 2. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

1620

1621 3.2.2.2.2 Feature Type "Setpoint"

1622 3.2.2.2.2.1 Function "setpointDescriptionListData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: R	measurementId	<m1#(1..1)>	SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. If the Use Case "Monitoring of Power Consumption" (MPC) is supported by the Actor Inverter, the same measurementId SHALL be used for <m1#1> of this Use Case and <m1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <m1#1> (in the Use Case MPC) is set to "18", <m1#1> (in this Use Case) SHALL be set to "18", too.
1: M	setpointType	"valueAbsolute"	
1: M	unit	"W"	
1: M	scopeType	"batteryAcPower"	
1: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
1: M	setpointId	<st2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointType	"valueAbsolute"	
1: M	unit	"W"	
1: M	scopeType	"batteryDcPower"	
2: M	Setpoint. setpointDescriptionListData. setpointDescriptionData.		
2: M	setpointId	<st3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	setpointType	"valueAbsolute"	
2: M	unit	"W"	
2: M	scopeType	"pccPower"	

1623 Table 33: Content of Function "setpointDescriptionListData" at Actor Inverter

1624

1625 3.2.2.2.2.2 Function "setpointConstraintsListData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/1]		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointRangeMin.		[COB-001/3], [COB-980/1/1]

			The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]
1: M	setpointRangeMin. number		SHALL be used.
1: M	setpointRangeMin. scale		SHALL be used.
1: M	setpointRangeMax.		[COB-001/3], [COB-980/1/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
1: M	setpointRangeMax. number		SHALL be used.
1: M	setpointRangeMax. scale		SHALL be used.
1: R	setpointStepSize.		[COB-001/3], [COB-980/1/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
1: M	setpointStepSize. number		SHALL be used.
1: M	setpointStepSize. scale		SHALL be used.
1: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/1]		
1: M	setpointId	<st2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	setpointRangeMin.		[COB-001/3], [COB-980/1/1] The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]
1: M	setpointRangeMin. number		SHALL be used.
1: M	setpointRangeMin. scale		SHALL be used.
1: M	setpointRangeMax.		[COB-001/3], [COB-980/1/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
1: M	setpointRangeMax. number		SHALL be used.
1: M	setpointRangeMax. scale		SHALL be used.
1: R	setpointStepSize.		[COB-001/3], [COB-980/1/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
1: M	setpointStepSize. number		SHALL be used.
1: M	setpointStepSize. scale		SHALL be used.
2: M	Setpoint. setpointConstraintsListData. setpointConstraintsData. [COB-980/2]		

2: M	setpointId	<st3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	setpointRangeMin.		[COB-001/3], [COB-980/2/1] The setpoint value SHALL NOT be smaller than setpointRangeMin. [Scaled number rules]
2: M	setpointRangeMin. number		SHALL be used.
2: M	setpointRangeMin. scale		SHALL be used.
2: M	setpointRangeMax.		[COB-001/3], [COB-980/2/2] The setpoint value SHALL NOT be larger than setpointRangeMax. [Scaled number rules]
2: M	setpointRangeMax. number		SHALL be used.
2: M	setpointRangeMax. scale		SHALL be used.
2: R	setpointStepSize.		[COB-001/3], [COB-980/2/3] SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server. [Scaled number rules]
2: M	setpointStepSize. number		SHALL be used.
2: M	setpointStepSize. scale		SHALL be used.

Table 34: Content of Function "setpointConstraintsListData" at Actor Inverter

3.2.2.2.3 Function "setpointListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Setpoint. setpointListData. setpointData.		
1: M	setpointId	<st1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[COB-001/1], [COB-011], [COB-960], [COB-980/1] [Scaled number rules]
1: M \C	value. number		SHALL be used.
1: M \C	value. scale		SHALL be used.
1: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
1: M \C	isSetpointActive	"true" "false"	[COB-004/1], [COB-006/1], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.

1: O, M \RW	timePeriod.		
1: O, M \RW	timePeriod. endTime		[COB-005/1]
1: M	Setpoint. setpointListData. setpointData.		
1: M	setpointId	<st2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[COB-001/1], [COB-012], [COB-960], [COB-980/1] [Scaled number rules]
1: M \C	value. number		SHALL be used.
1: M \C	value. scale		SHALL be used.
1: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
1: M \C	isSetpointActive	"true" "false"	[COB-004/1], [COB-006/1], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
1: O, M \RW	timePeriod.		
1: O, M \RW	timePeriod. endTime		[COB-005/1]
2: M	Setpoint. setpointListData. setpointData.		
2: M	setpointId	<st3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/1], [COB-021], [COB-960], [COB-980/2] [Scaled number rules]
2: M \C	value. number		SHALL be used.
2: M \C	value. scale		SHALL be used.
2: M	isSetpointChangeable	"true"	If set to "true", the value, isSetpointActive and timePeriod Elements SHALL be writeable by a client.
2: M \C	isSetpointActive	"true" "false"	[COB-004/2], [COB-006/2], [COB-009] If set to "false", the setpoint and its timePeriod and value Elements SHALL be ignored. If set to "true", the timePeriod and value Elements SHALL be applied.
2: O, M \RW	timePeriod.		
2: O, M \RW	timePeriod. endTime		[COB-005/2]

Table 35: Content of Function "setpointListData" at Actor Inverter

3.2.2.2.3 Feature Type "DeviceConfiguration"

3.2.2.2.3.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		

2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxAcChargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxAcDischargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxDcChargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"maxDcDischargePower"	
2: M	valueType	"scaledNumber"	
2: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"batteryActiveControlMode"	
3: M	valueType	"string"	
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultAcPower"	
3: M	valueType	"scaledNumber"	
3: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultDcPower"	
3: M	valueType	"scaledNumber"	

3: M	unit	"W"	The unit SHALL be applied to the value of the key.
3: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
3: M	keyId	<k8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	keyName	"defaultPccPower"	
3: M	valueType	"scaledNumber"	
3: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeAcPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeDcPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafePccPowerSetpoint"	
4: M	valueType	"scaledNumber"	
4: M	unit	"W"	The unit SHALL be applied to the value of the key.
4: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
4: M	keyId	<k12#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	keyName	"failsafeDurationMinimum"	
4: M	valueType	"duration"	

Table 36: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Inverter

3.2.2.2.3.2 Function "deviceConfigurationKeyValueListData"

Scenario [...]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
-----------------------------------	---------	-------	---

2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-022] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-023] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-024] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \C	value. scaledNumber. number		SHALL be used.
2: M \C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).

2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[COB-001/2], [COB-025] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
2: M \\C	value. scaledNumber. number		SHALL be used.
2: M \\C	value. scaledNumber. scale		SHALL be used.
2: M	isValueChangeable	"true"	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-031] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M \\C	value. string	"power"	[COB-031/1]
		"pcc"	[COB-031/2]
		"auto"	[COB-031/3]
3: M	isValueChangeable	"true" "false"	[COB-030/1] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-001/1], [COB-032] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2] If set to "false" the server SHALL decline change requests.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		

3: M	keyId	<k7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-001/1], [COB-033] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2] If set to "false" the server SHALL decline change requests.
3: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
3: M	keyId	<k8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	value.		[COB-001/1], [COB-034] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
3: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
3: M	value. scaledNumber. number		SHALL be used.
3: M	value. scaledNumber. scale		SHALL be used.
3: M	isValueChangeable	"false"	[COB-030/2] If set to "false" the server SHALL decline change requests.
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-041] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \C	value. scaledNumber. number		SHALL be used.
4: M \C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		

4: M	keyId	<k10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-042] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \\C	value. scaledNumber. number		SHALL be used.
4: M \\C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-001/1], [COB-043] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
4: M \\C	value. scaledNumber. number		SHALL be used.
4: M \\C	value. scaledNumber. scale		SHALL be used.
4: M	isValueChangeable	"true"	[COB-040] If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client).
4: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
4: M	keyId	<k12#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	value.		[COB-044] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
4: M \\C	value. duration	[2h..24h]* ¹	
4: M	isValueChangeable	"true"	[COB-044/2] If set to "true" the server SHALL evaluate changes of the element "value" by appropriate clients (e.g. the bound client) according to rules stated in [COB-044/4].

1636

Table 37: Content of Function "deviceConfigurationKeyValueListData" at Actor Inverter

*¹: Values within the duration Element can be represented in different ways. E.g.: "2 hours" can be represented as "PT2H" or "PT120M".

3.2.2.2.4 Feature Type "DeviceDiagnosis"

3.2.2.2.4.1 Function "deviceDiagnosisHeartbeatData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
5: M	timestamp		SHALL hold the time of creation.
5: M	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
5: M	heartbeatTimeout	≤60 seconds* ¹ , * ² [COB-008]	[COB-052] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 38: Content of Function "deviceDiagnosisHeartbeatData" at Actor Inverter

*¹: Could be e.g. "PT60S" or "PT1M"

*²: For the COB Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

3.2.2.3 Client data - Specializations

3.2.2.3.1 Topic "DeviceDiagnosisHeartbeat"

3.2.2.3.1.1 Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds"

Scenario {...}: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	DeviceDiagnosis.deviceDiagnosisHeartbeatData.		
5: M	timestamp		SHALL hold the time of creation.
5: R* ³	heartbeatCounter		The value of the heartbeatCounter Element SHALL be increased after every heartbeatTimeout (increase with every heartbeat notification message). The deviceDiagnosisHeartbeatData Function can not only be notified by the device due to a subscription, but can be requested with a read request by another device, too. In this case the device sends out a reply message. If the device sends out a reply message the Element heartbeatCounter SHALL NOT be incremented and the heartbeatTimeout has its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).
5: R* ³	heartbeatTimeout	≤60 seconds* ¹ , * ² [COB-007]	[COB-051] deviceDiagnosisHeartbeatData SHALL be notified at least each heartbeatTimeout period.

Table 39: Content of Specialization "DeviceDiagnosisHeartbeat_Timeout60Seconds" at Actor Inverter

*¹: Could be e.g. "PT60S" or "PT1M"

*²: For the COB Use Case the heartbeat timeout is 60 seconds. As this heartbeat functionality may also be used within other parallel Use Cases with shorter timeouts, the Element heartbeatTimeout SHALL be ≤60 seconds.

*³: The Inverter is not obliged to verify this Element within the CEM's Heartbeat message.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- Detailed discovery: allows to discover resource addresses.
- Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.

1666 c) Subscription: allows to subscribe to resource addresses, which is necessary to receive
1667 unsolicited notifications if a resource changes during runtime.

1668 It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- 1669 - E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription
1670 needs to occur.
- 1671 - E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs
1672 to occur only once for all Use Cases.

1673 Depending on which Entity, Feature and Functions are used within a Scenario the payload of the
1674 corresponding messages may slightly differ, but the basic principles and messages used stay the
1675 same.

1676 The subsequent messages SHALL be exchanged for those parts that have not already been performed
1677 since the current connection is established or if those parts are outdated for another reason (e.g. if
1678 the detailed discovery is needed, but the bindings and subscriptions are still active from a previous
1679 connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario
1680 communication parts are up to date, this section MAY be skipped, and the implementation can
1681 proceed as described in the corresponding "Scenario communication" sections.

1682 After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-
1683 Scenario communication SHALL be performed as well. There may be circumstances where messages
1684 from the Pre-Scenario communication may be exchanged again.

1685 Often the necessary messages of different Scenarios can be combined, so that only one single
1686 message is needed instead of multiple messages for the different Scenarios. This also is the case for
1687 the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery
1688 is necessary, as well as only one subscription request or binding request is needed for each Feature.
1689 Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or
1690 binding is necessary.

1691

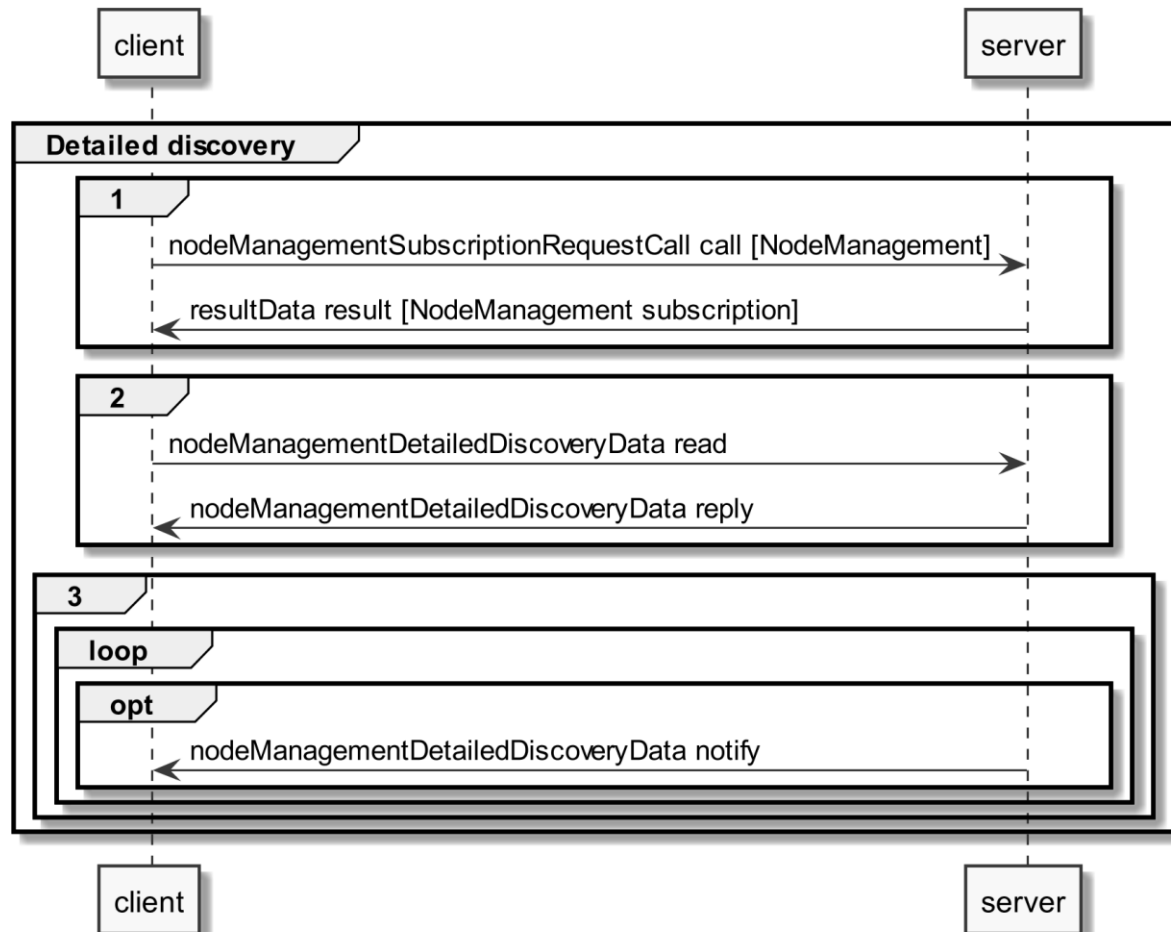
1692 3.3.2 Detailed discovery

1693 For the functionality where a client already has current detailed discovery information (i.e.
1694 independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

1695 Otherwise, the following procedure SHALL be performed in the given order:

- 1696 1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL
1697 acquire a subscription according to the figure provided within this sub-section.
- 1698 2. A client SHALL read the detailed discovery information according to the figure provided
1699 within this sub-section. It SHALL keep the received information as far as it concerns
1700 mandatory and supported optional Entity Types, Feature Types and Functions of this Use
1701 Case that are needed by the client. This means that a client may choose how to store the
1702 necessary information. E.g. a client Actor can store the information how to address the
1703 necessary Features of the implemented Scenarios but may discard the Entity information.

- 1704 3. If and as long as a client has a subscription to the detailed discovery information of an Actor
 1705 and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed
 1706 discovery information) the received information as far as it concerns mandatory and
 1707 supported optional Entity Types, Feature Types and Functions of this Use Case.



1708

1709 *Figure 8: Pre-Scenario communication - Detailed discovery sequence diagram*

1710 If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the
 1711 detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message
 1712 was received successfully.

1713 If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply"
 1714 message contains at least the mandatory Entities and Features given in the "Actor [...] overview"
 1715 diagrams as well as the used Functions and their "possible operations" described in section 3.2 and
 1716 its sub-sections.

1717 If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData
 1718 reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...]
 1719 overview" diagrams as well as all needed Functions and their "possible operations" described in
 1720 section 3.2 and its sub-sections. However, as soon as the functionality is available it will be
 1721 announced via a "nodeManagementDetailedDiscoveryData notify" message.

1722 For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial
 1723 read with separated Selectors that may use one of the following Elements:

- 1724 - entityType
- 1725 - featureType

1726 Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case
 1727 may be discovered. However, only Features and Entities that fulfil the hierarchical order as described
 1728 within the Actors' sections shall be considered for this Use Case.

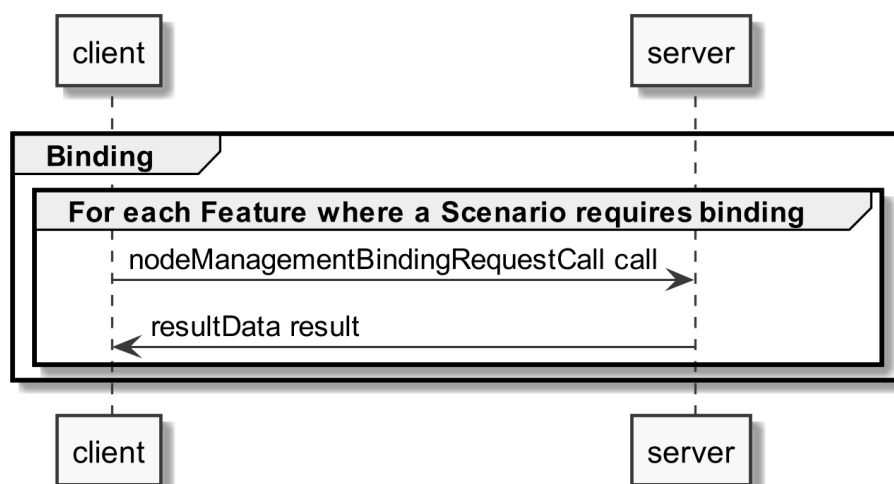
1729 A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify
 1730 SHOULD also be supported with separated Selectors that may use one of the following Elements:

- 1731 - entityAddress
- 1732 - featureAddress

1733

1734 3.3.3 Binding

1735 If binding is required by a Scenario that uses Features with writeable or changeable data, the server
 1736 SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs,
 1737 the client SHALL create a binding to the corresponding Feature. For this the
 1738 nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:



1739

1740 *Figure 9: Pre-Scenario communication - Binding sequence diagram*

1741 If functionality is added or removed dynamically, binding may not be possible at all times on the
 1742 required Functions. A client SHALL retry to create a binding again when receiving according updated
 1743 detailed discovery information.

1744

1745 3.3.4 Subscription

1746 A server SHALL support subscription for all Features that contain readable data that may change
 1747 during runtime. The client SHALL create a subscription for all Features that the client wants to read.
 1748 For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following
 1749 sequence diagram:

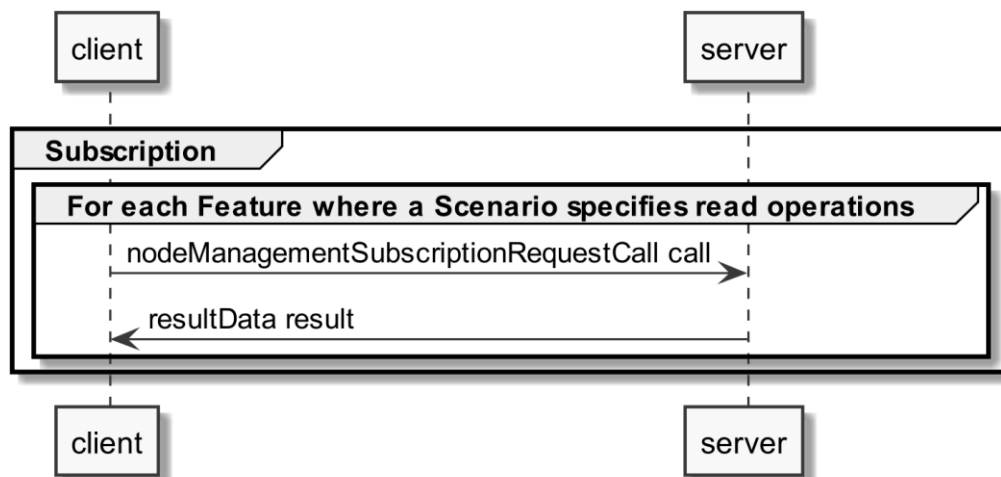


Figure 10: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Control mode "Power"

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

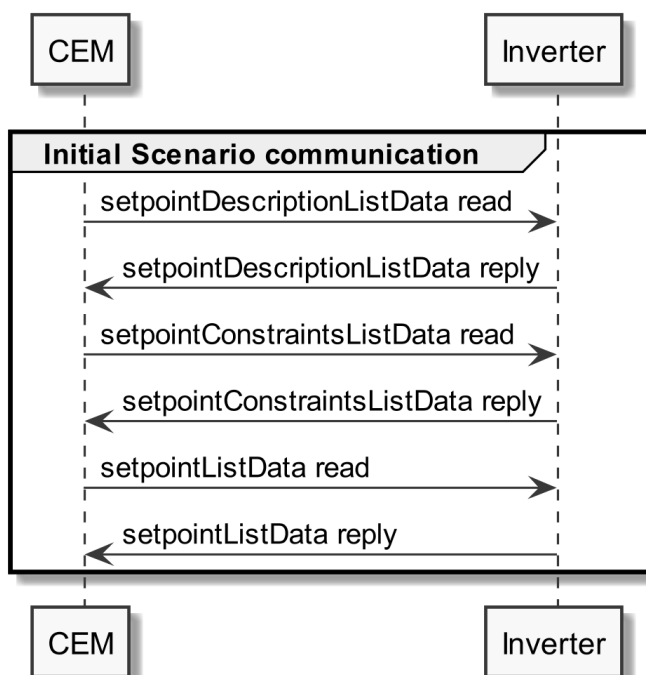
1778 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
 1779 are known and the necessary binding and subscription procedures have been finished. However, as
 1780 soon as the address of a required resource is known, the Initial Scenario communication for this
 1781 resource MAY start already, even if the addresses of other required resources are not known yet.

1782 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1783 communication is triggered again for those resources.

1784

1785 **3.4.1.2 Initial Scenario communication**

1786 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1787 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1788 resources may have changed in the meantime:



1789

1790 *Figure 11: Scenario 1 - Initial Scenario communication sequence diagram*

1791 **Partial data information regarding [COB-011]:**

1792 The setpointDescriptionListData read SHOULD be a "partial" read operation with the following
 1793 Selector:

- 1794 - scopeType = "batteryAcPower"

1795 The setpointConstraintsListData read and setpointListData read SHOULD be "partial" read operations
 1796 with the following Selector:

- 1797 - setpointId (derived from the setpointDescriptionListData reply)

1798 Note: If partial read is not supported, a full read SHALL be performed.

1799

1800 **Partial data information regarding [COB-012]:**

1801 The setpointDescriptionListData read SHOULD be a "partial" read operation with the following
1802 Selector:

1803 - scopeType = "batteryDcPower"

1804 The setpointConstraintsListData read and setpointListData read SHOULD be "partial" read operations
1805 with the following Selector:

1806 - setpointId (derived from the setpointDescriptionListData reply)

1807 Note: If partial read is not supported, a full read SHALL be performed.

1808

1809 The following table shows where the required content of the messages from the sequence diagram is
1810 described:

Message name from sequence diagram	Content description in table	Scenario number in table
setpointDescriptionListData reply	Table 33	1
setpointConstraintsListData reply	Table 34	1
setpointListData reply	Table 35	1

1811 *Table 40: Initial Scenario communication content references for Scenario 1*

1812 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1813 provided completely, but later during Runtime Scenario communication.

1814

1815 **3.4.1.3 Runtime Scenario communication**

1816 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1817 during runtime.

1818 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
1819 in the following figure:

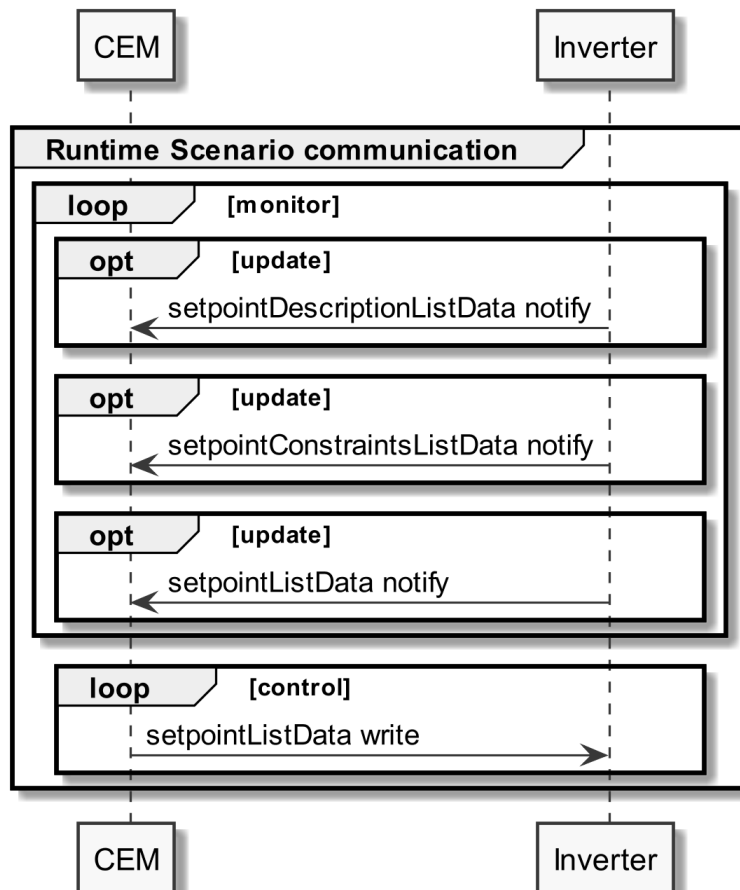


Figure 12: Scenario 1 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "setpointListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For setpointDescriptionListData notify, setpointConstraintsListData notify and setpointListData notify "partial" delete notifications SHOULD be supported with the Selector:

- setpointId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table

setpointDescriptionListData notify	Table 33	1
setpointConstraintsListData notify	Table 34	1
setpointListData notify	Table 35	1
setpointListData write	Table 35	1

Table 41: Runtime Scenario communication content references for Scenario 1

3.4.1.4 Additional information

If upon receipt of a write message the setpoint cannot be applied by the Inverter, the Inverter SHALL respond with an "Application error" message (NACK, [COB-003/1]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).

Special care must be taken by the Actor CEM in case a previous setpoint WITH the Element endTime set is overwritten with a new setpoint WITHOUT the Element endTime. Due to the "partial" concept, where only those Elements are overwritten that are set within the write command, the old endTime Element would remain valid. In such a case, the CEM has to delete the old endTime Element with a partial delete command (that can be part of the write command that includes the new setpoint). The following example shows this combined command:

```

...
<payload>
  <cmd>
    <function>setpointListData</function>
    <filter>
      <cmdControl><delete/></cmdControl>
      <setpointListDataSelectors>
        <setpointId>(*1)</setpointId>
      </setpointListDataSelectors>
      <setpointDataElements>
        <timePeriod>
          <endTime/>
        </timePeriod>
      </setpointDataElements>
    </filter>
    <filter>
      <cmdControl><partial/></cmdControl>
    </filter>
    <setpointListData>
      (*2)
    </setpointListData>
  </cmd>
</payload>

```

where "(*1)" must be replaced by the proper value of "setpointId" and "(*2)" must contain a proper "setpointListData" instance with the same value of "setpointId".

Note: Setting always all changeable or writeable data in (*2) can help to avoid race conditions as described at [COB-010/1].

1877 **3.4.2 Scenario 2 - Control mode "PCC"**

1878 **3.4.2.1 Pre-Scenario communication**

- 1879 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
1880 of the server Features used in the Initial Scenario communication. If the address of a
1881 particular server Feature is not known, the detailed discovery must be used, as described in
1882 section 3.3.2.
- 1883 2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as
1884 described in section 3.3.3. Only one binding partner is allowed to write the data specified in
1885 this Scenario.
- 1886 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
1887 the corresponding Actor within this Scenario, as described in section 3.3.4.

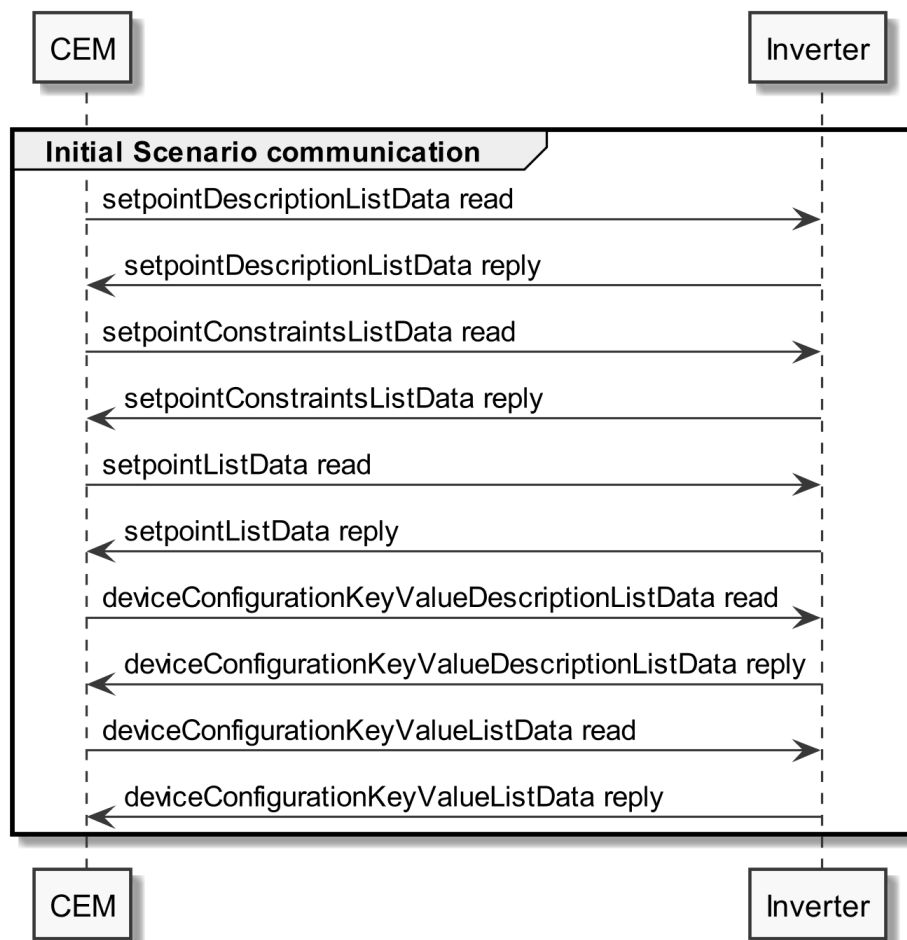
1888 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
1889 are known and the necessary binding and subscription procedures have been finished. However, as
1890 soon as the address of a required resource is known, the Initial Scenario communication for this
1891 resource MAY start already, even if the addresses of other required resources are not known yet.

1892 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1893 communication is triggered again for those resources.

1894

1895 **3.4.2.2 Initial Scenario communication**

1896 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
1897 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
1898 resources may have changed in the meantime:



1899

1900 *Figure 13: Scenario 2 - Initial Scenario communication sequence diagram*1901 **Partial data information regarding [COB-021]:**

1902 The setpointDescriptionListData read SHOULD be a "partial" read operation with the following
 1903 Selector:

- 1904 - scopeType = "pccPower"

1905 The setpointConstraintsListData read and setpointListData read SHOULD be "partial" read operations
 1906 with the following Selector:

- 1907 - setpointId (derived from the setpointDescriptionListData reply)

1908 Note: If partial read is not supported, a full read SHALL be performed.

1909

1910 **Partial data information regarding [COB-022]:**

1911 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
 1912 with the following Selector:

- 1913 - keyName = "maxAcChargePower"

1914 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
1915 following Selector:

- 1916 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1917 Note: If partial read is not supported, a full read SHALL be performed.

1918

1919 **Partial data information regarding [COB-023]:**

1920 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1921 with the following Selector:

- 1922 - keyName = "maxAcDischargePower"

1923 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
1924 following Selector:

- 1925 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1926 Note: If partial read is not supported, a full read SHALL be performed.

1927

1928 **Partial data information regarding [COB-024]:**

1929 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1930 with the following Selector:

- 1931 - keyName = "maxDcChargePower"

1932 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
1933 following Selector:

- 1934 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1935 Note: If partial read is not supported, a full read SHALL be performed.

1936

1937 **Partial data information regarding [COB-025]:**

1938 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1939 with the following Selector:

- 1940 - keyName = "maxDcDischargePower"

1941 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
1942 following Selector:

- 1943 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1944 Note: If partial read is not supported, a full read SHALL be performed.

1945

1946 The following table shows where the required content of the messages from the sequence diagram is
 1947 described:

Message name from sequence diagram	Content description in table	Scenario number in table
setpointDescriptionListData reply	Table 33	2
setpointConstraintsListData reply	Table 34	2
setpointListData reply	Table 35	2
deviceConfigurationKeyValueDescriptionListData reply	Table 36	2
deviceConfigurationKeyValueListData reply	Table 37	2

1948 *Table 42: Initial Scenario communication content references for Scenario 2*

1949 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1950 provided completely, but later during Runtime Scenario communication.

1951

1952 **3.4.2.3 Runtime Scenario communication**

1953 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1954 during runtime.

1955 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1956 in the following figure:

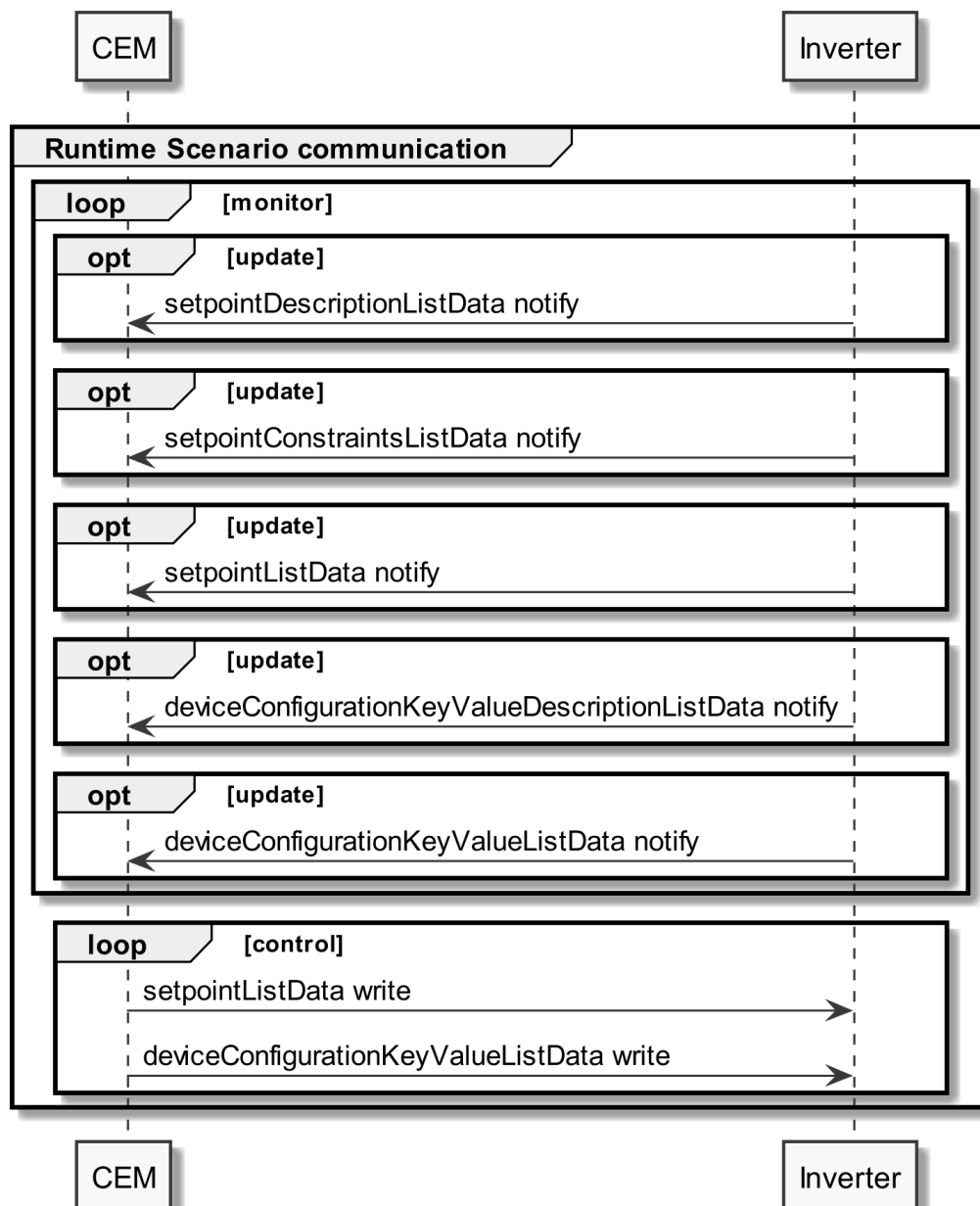


Figure 14: Scenario 2 - Runtime Scenario communication sequence diagram

For Setpoint related Functions:

Note: Normally, in this Scenario only the "setpointListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For setpointDescriptionListData notify, setpointConstraintsListData notify and setpointListData notify "partial" delete notifications SHOULD be supported with the Selector:

- setpointId

For DeviceConfiguration related Functions:

1969 Note: Normally, in this Scenario only the "deviceConfigurationKeyValueListData" Function changes
 1970 during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during
 1971 runtime.

1972 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1973 Scenario.

1974 For deviceConfigurationKeyValueDescriptionListData notify and
 1975 deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with
 1976 the Selector:

1977 - keyId

1978

1979 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1980 Scenario communication phase is required.

1981 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1982 not be evaluated.

1983

1984 The following table shows where the required content of the messages of the sequence diagram is
 1985 described:

Message name from sequence diagram	Content description in table	Scenario number in table
setpointDescriptionListData notify	Table 33	2
setpointConstraintsListData notify	Table 34	2
setpointListData notify	Table 35	2
deviceConfigurationKeyValueDescriptionListData notify	Table 36	2
deviceConfigurationKeyValueListData notify	Table 37	2
setpointListData write	Table 35	2
deviceConfigurationKeyValueListData write	Table 37	2

1986 *Table 43: Runtime Scenario communication content references for Scenario 2*

1987

1988 **3.4.2.4 Additional information**

1989 If upon receipt of a write message the setpoint cannot be applied by the Inverter, the Inverter SHALL
 1990 respond with an "Application error" message (NACK, [COB-003/1]) using the "result" classifier and
 1991 resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in
 1992 [ProtocolSpecification]).

1993 Special care must be taken by the Actor CEM in case a previous setpoint WITH the Element endTime
 1994 set is overwritten with a new setpoint WITHOUT the Element endTime. Due to the "partial" concept,
 1995 where only those Elements are overwritten that are set within the write command, the old endTime
 1996 Element would remain valid. In such a case, the CEM has to delete the old endTime Element with a

1997 partial delete command (that can be part of the write command that includes the new setpoint). The
 1998 following example shows this combined command:

```

1999     ...
2000     <payload>
2001         <cmd>
2002             <function>setpointListData</function>
2003             <filter>
2004                 <cmdControl><delete/></cmdControl>
2005                 <setpointListDataSelectors>
2006                     <setpointId>(*1)</setpointId>
2007                 </setpointListDataSelectors>
2008                 <setpointDataElements>
2009                     <timePeriod>
2010                         <endTime/>
2011                     </timePeriod>
2012                 </setpointDataElements>
2013             </filter>
2014             <filter>
2015                 <cmdControl><partial/></cmdControl>
2016             </filter>
2017             <setpointListData>
2018                 (*2)
2019             </setpointListData>
2020         </cmd>
2021     </payload>

```

2022 where "(*1)" must be replaced by the proper value of "setpointId" and "(*2)" must contain a proper
 2023 "setpointListData" instance with the same value of "setpointId".

2024 Note: Setting always all changeable or writeable data in (*2) can help to avoid race conditions as
 2025 described at [COB-010/2].

2026

2027 3.4.3 Scenario 3 - Configuration parameters

2028 3.4.3.1 Pre-Scenario communication

- 2029 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
 2030 of the server Features used in the Initial Scenario communication. If the address of a
 2031 particular server Feature is not known, the detailed discovery must be used, as described in
 2032 section 3.3.2.
- 2033 2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as
 2034 described in section 3.3.3. Only one binding partner is allowed to write the data specified in
 2035 this Scenario.
- 2036 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
 2037 the corresponding Actor within this Scenario, as described in section 3.3.4.

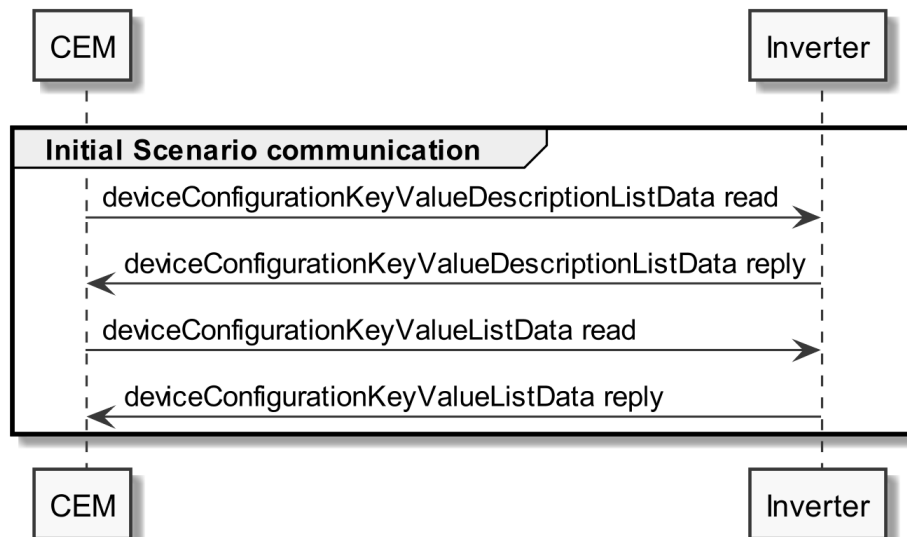
2038 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
 2039 are known and the necessary binding and subscription procedures have been finished. However, as
 2040 soon as the address of a required resource is known, the Initial Scenario communication for this
 2041 resource MAY start already, even if the addresses of other required resources are not known yet.

2042 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 2043 communication is triggered again for those resources.

2044

2045 **3.4.3.2 Initial Scenario communication**

2046 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 2047 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 2048 resources may have changed in the meantime:



2049

2050 *Figure 15: Scenario 3 - Initial Scenario communication sequence diagram*2051 **Partial data information regarding [COB-031]:**

2052 The `deviceConfigurationKeyValueDescriptionListData read` SHOULD be a "partial" read operation
 2053 with the following Selector:

- 2054 - `keyName = "batteryActiveControlMode"`

2055 The `deviceConfigurationKeyValueListData read` SHOULD be "partial" read operation with the
 2056 following Selector:

- 2057 - `keyId` (derived from the `deviceConfigurationKeyValueDescriptionListData reply`)

2058 Note: If partial read is not supported, a full read SHALL be performed.

2059

2060 **Partial data information regarding [COB-032]:**

2061 The `deviceConfigurationKeyValueDescriptionListData read` SHOULD be a "partial" read operation
 2062 with the following Selector:

- 2063 - `keyName = "defaultAcPower"`

2064 The `deviceConfigurationKeyValueListData read` SHOULD be "partial" read operation with the
 2065 following Selector:

- 2066 - `keyId` (derived from the `deviceConfigurationKeyValueDescriptionListData reply`)

2067 Note: If partial read is not supported, a full read SHALL be performed.

2068

2069 **Partial data information regarding [COB-033]:**

2070 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
2071 with the following Selector:

2072 - keyName = "defaultDcPower"

2073 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
2074 following Selector:

2075 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

2076 Note: If partial read is not supported, a full read SHALL be performed.

2077

2078 **Partial data information regarding [COB-034]:**

2079 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
2080 with the following Selector:

2081 - keyName = "defaultPccPower"

2082 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
2083 following Selector:

2084 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

2085 Note: If partial read is not supported, a full read SHALL be performed.

2086

2087 The following table shows where the required content of the messages from the sequence diagram is
2088 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData reply	Table 36	3
deviceConfigurationKeyValueListData reply	Table 37	3

2089 *Table 44: Initial Scenario communication content references for Scenario 3*

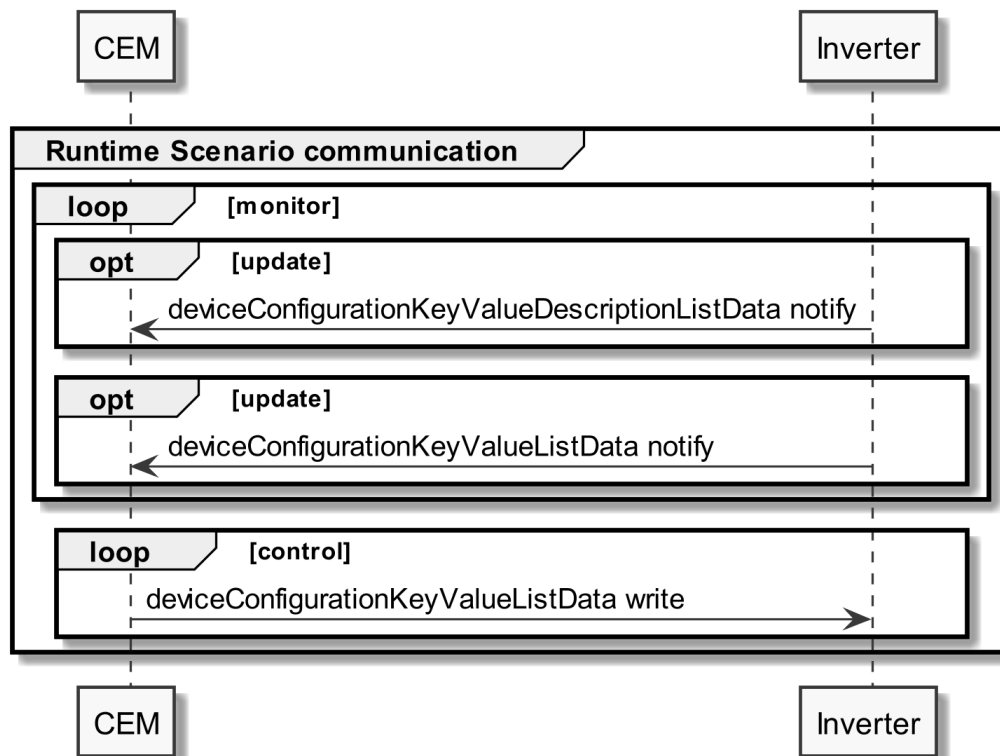
2090 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
2091 provided completely, but later during Runtime Scenario communication.

2092

2093 **3.4.3.3 Runtime Scenario communication**

2094 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
2095 during runtime.

2096 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 2097 in the following figure:



2098

2099 *Figure 16: Scenario 3 - Runtime Scenario communication sequence diagram*

2100 Note: Normally, in this Scenario only the "deviceConfigurationKeyValueListData" Function changes
 2101 during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during
 2102 runtime.

2103 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 2104 Scenario.

2105 For deviceConfigurationKeyValueDescriptionListData notify and
 2106 deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with
 2107 the Selector:

2108 - keyId

2109 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 2110 Scenario communication phase is required.

2111 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 2112 not be evaluated.

2113

2114 The following table shows where the required content of the messages of the sequence diagram is
 2115 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData notify	Table 36	3
deviceConfigurationKeyValueListData notify	Table 37	3
deviceConfigurationKeyValueListData write	Table 37	3

Table 45: Runtime Scenario communication content references for Scenario 3

3.4.3.4 Additional information

If upon receipt of a write message the configuration parameter cannot be applied by the Inverter, the Inverter SHALL respond with an "Application error" message (NACK, [COB-003/2]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).

3.4.4 Scenario 4 - Failsafe values

3.4.4.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.4.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

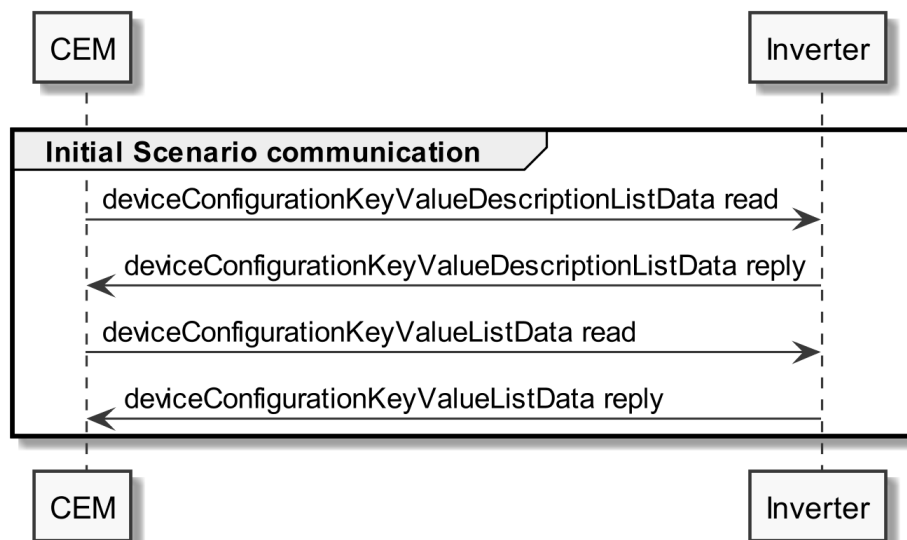


Figure 17: Scenario 4 - Initial Scenario communication sequence diagram

Partial data information regarding [COB-041]:

The `deviceConfigurationKeyValueDescriptionListData` read SHOULD be a "partial" read operation with the following Selector:

- `keyName = "failsafeAcPowerSetpoint"`

The `deviceConfigurationKeyValueListData` read SHOULD be "partial" read operation with the following Selector:

- `keyId` (derived from the `deviceConfigurationKeyValueDescriptionListData` reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [COB-042]:

The `deviceConfigurationKeyValueDescriptionListData` read SHOULD be a "partial" read operation with the following Selector:

- `keyName = "failsafeDcPowerSetpoint"`

The `deviceConfigurationKeyValueListData` read SHOULD be "partial" read operation with the following Selector:

- `keyId` (derived from the `deviceConfigurationKeyValueDescriptionListData` reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [COB-043]:

The `deviceConfigurationKeyValueDescriptionListData` read SHOULD be a "partial" read operation with the following Selector:

2169 - keyName = "failsafePccPowerSetpoint"

2170 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
2171 following Selector:

2172 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

2173 Note: If partial read is not supported, a full read SHALL be performed.

2174

2175 **Partial data information regarding [COB-044]:**

2176 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
2177 with the following Selector:

2178 - keyName = "failsafeDurationMinimum"

2179 The deviceConfigurationKeyValueListData read SHOULD be "partial" read operation with the
2180 following Selector:

2181 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

2182 Note: If partial read is not supported, a full read SHALL be performed.

2183

2184 The following table shows where the required content of the messages from the sequence diagram is
2185 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData reply	Table 36	4
deviceConfigurationKeyValueListData reply	Table 37	4

2186 *Table 46: Initial Scenario communication content references for Scenario 4*

2187 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
2188 provided completely, but later during Runtime Scenario communication.

2189

2190 **3.4.4.3 Runtime Scenario communication**

2191 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
2192 during runtime.

2193 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
2194 in the following figure:

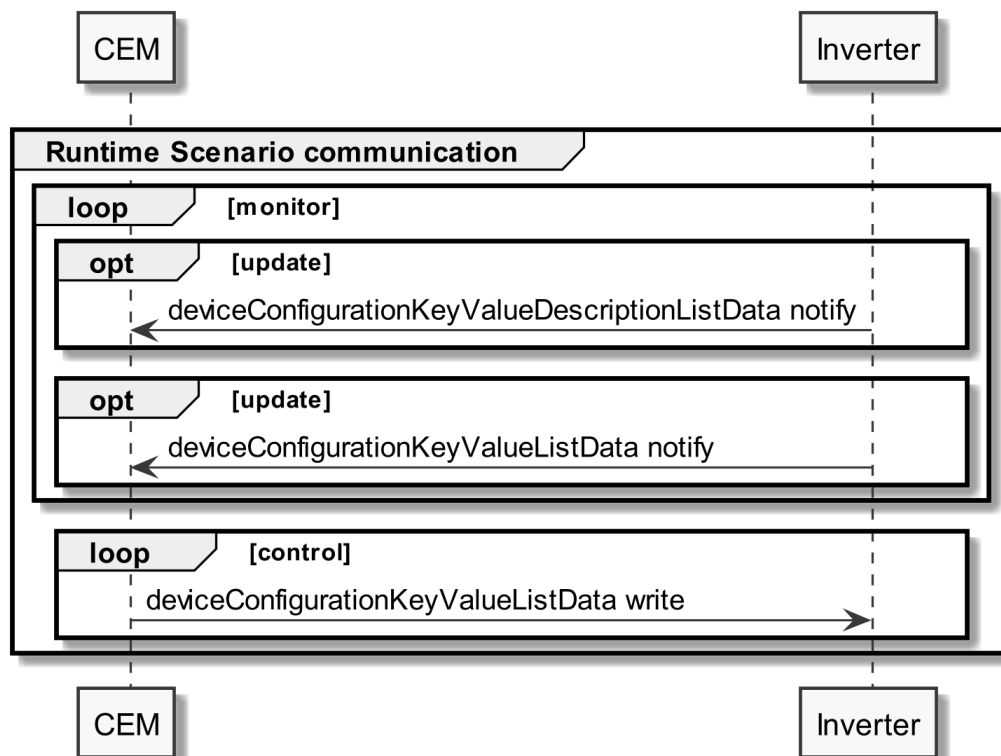


Figure 18: Scenario 4 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "deviceConfigurationKeyValueListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For deviceConfigurationKeyValueDescriptionListData notify and deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with the Selector:

- keyId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData notify	Table 36	4

deviceConfigurationKeyValueListData notify	Table 37	4
deviceConfigurationKeyValueListData write	Table 37	4

Table 47: Runtime Scenario communication content references for Scenario 3

3.4.4.4 Additional information

If a received write command on a failsafe value cannot be applied by the Inverter, the Inverter SHALL respond with an "Application error" message (NACK, [COB-003/3]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).

Note: In case [COB-044/5] is applied and there was a change of the Inverter's Failsafe Duration Minimum's value, the new value is notified to the CEM. If there was no change (because the Failsafe Duration Minimum was already set to the Inverter's maximum value for the Failsafe Duration Minimum), there will be no notification.

3.4.5 Scenario 5 - Heartbeat

3.4.5.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding SHOULD NOT be used for this Scenario.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.5.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

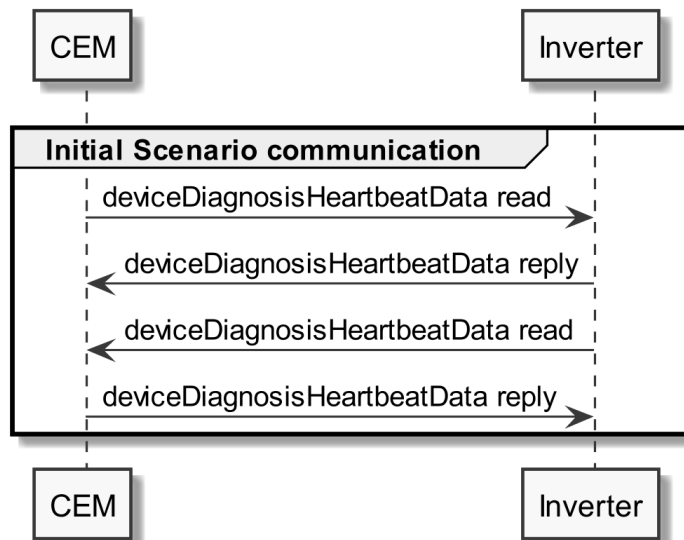


Figure 19: Scenario 5 - Initial Scenario communication sequence diagram

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceDiagnosisHeartbeatData reply (of Actor CEM)	Table 14	5
deviceDiagnosisHeartbeatData reply (of Actor Inverter)	Table 38	5

Table 48: Initial Scenario communication content references for Scenario 5

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.5.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

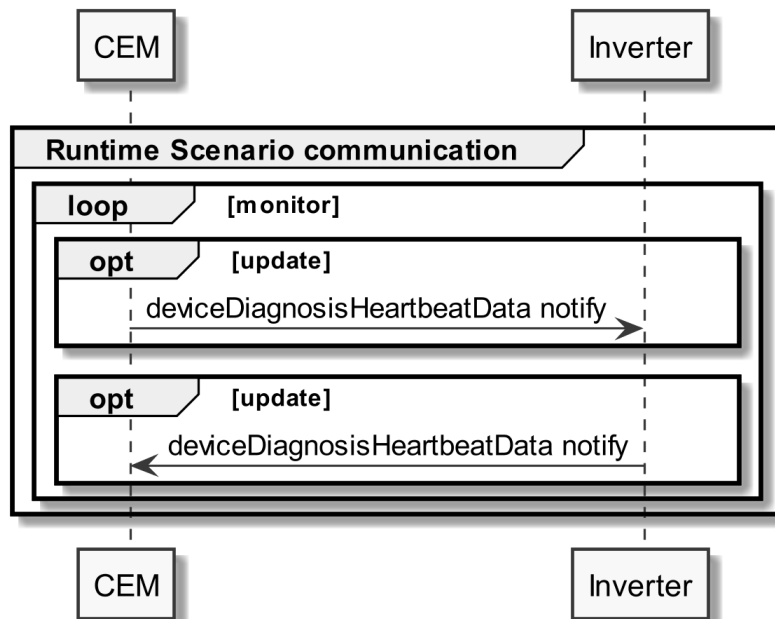


Figure 20: Scenario 5 - Runtime Scenario communication sequence diagram

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceDiagnosisHeartbeatData notify (of Actor CEM)	Table 14	5
deviceDiagnosisHeartbeatData notify (of Actor Inverter)	Table 38	5

Table 49: Runtime Scenario communication content references for Scenario 5

3.4.5.4 Additional information

None.